

Manual

Special Input/Information Functions for Material, Batches, Serial Numbers AIP-LCS 8.1

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1 Special Data Collection Functions for Batches and Serial Numbers

Purpose

The AIP features provided in this function package make it possible to enter batch-related data directly in production using shop floor terminals or data acquisition PCs.

Integration

Data entered via the AIP terminal can be displayed and/or evaluated in different MOC applications. Collected data can be uploaded via interfaces.

Features

Besides batch data, HYDRA shop floor clients (AIP) also collect and generate serial numbers to provide for complete documentation and traceability. The function package additionally provides further special features for palletizing, packaging, assembling/finishing and weighing of components.

Functions to collect and use serial numbers:

- Predefined series or serial numbers are transferred in HYDRA standard format via the HYDRA
- ERP interface
- Serial numbers are entered or generated via a user-friendly posting function at the BDE terminal
- Option to assign components also identified by serial numbers
- Classification into yield/ scrap quantities with reasons
- Automatic generation of goods receipts/ goods issues
- Optional verification of serial numbers already defined for the order
- Entry of order-related series and serial numbers identifying processed parts
- Validation check for already posted series or serial numbers
- Display of available and already used series or serial numbers on workstation PCs (MES Operation Center)
- Upload of posted series and serial numbers in HYDRA standard format via the HYDRA ERP interface

Functions for palletizing / packaging / assembling/finishing:

- Mapping of specific data collection processes for palletizing and packaging at packing stations

- Generation of handling units at the BDE terminal
- Function to edit handling units (e.g. assigning batches and removing batches from handling units)

Functions to weigh components:

- Display of the components to be weighed
- Management of multiple charges for each weighing order
- Presentation of target / actual quantities for each component and charge
- Weight recording via the interface connected to the scale
- Warning if tolerance quantities are exceeded or not reached
- Function to correct quantities and/or to change charge quantities for components
- Validation check for weighed batches

Additional licenses may be needed in order to use the functions listed above. Adding and coordinating the specific requirements and implementing them are considered a customized HYDRA service.

2 Operation of AIP

2.1 Special control and display elements within AIP

Tables

Tables are displayed in a uniform way within AIP. This affects the basic display (workplaces, operations, ...) as well as the selection lists of posting dialogs.

Filter	
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE

Page 1 • Page 2

Provided that information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. By clicking/touching the user can directly switch to another page.

An operation may be selected using the mouse, touch screen, keyboard (arrow keys: '↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Please find the following example: When an operation is logged on, those operations may be selected that are included in the sequencing list or that are planned for the corresponding workplace or group. However, when operations are interrupted, only running operations may be selected.



Scrolling page by page (up or down) in the table.



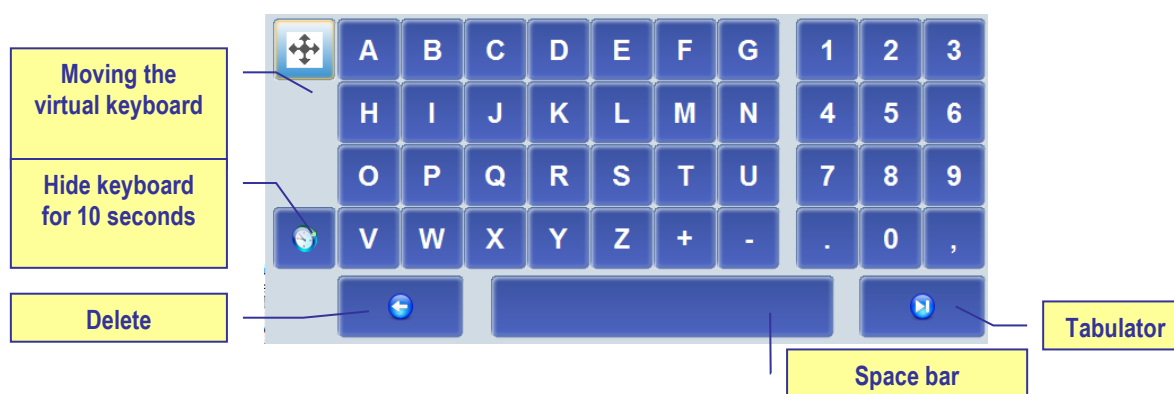
Scrolling to the left or right. Only those buttons are activated that are reasonable for the current situation. This figure shows that scrolling to the left has been deactivated.

Filter

A “table filter” may optionally be displayed (customizing). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is case-insensitive.

Virtual keyboard

The virtual keyboard allows for data to be entered manually via touch screen or a connected mouse. To make it easier for inexperienced users to find the required keys, the numeric key pad is organized like the telephone and letters are aligned in alphabetical order. Consequently, both differ from the computer keyboard which usually is aligned in the “QWERTZ keyboard layout”. The virtual keyboard is displayed automatically as soon as an input field is focused.



The driver needs to be configured respectively for the touch screen to be able to move the keyboard (settings in the control panel of the terminal/PC)! The virtual keyboard only supports the characters "0" - "9", "A" - "Z" and "+", "-", ".", and ",". Other characters or languages are not supported. It is recommendable to use an additional keyboard if texts in other languages have to be entered.

The start position of the virtual keyboard can be defined by a setting in the configuration file keyboard.ini. Subject to the screen resolution, the parameters xpos= and ypos= need to be enabled in the configuration file.

If the virtual keyboard is not to be shown in general, the parameter –t needs to be included in the parameter bar parameters= of the configuration file ctaip.ini.

Date display

AIP supports a country-specific date format in dynamic dialogs. This can be configured in the "control panel", "regional settings", "short date" dialog of the terminal/PC. The following has to be taken into account in this context:

- Years are always four characters long.
- Months and days are always 2 characters long.
- “-”, “/” and “.” are allowed separators
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- English(USA) MM/DD/YYYY
- Danish MM-DD-YYYY
- Customer-specific 1 YYYY-MM-DD
- Customer-specific 2 YYYY/MM/DD
- Customer-specific 3 MM/YYYY/DD

Please note

If the date format does not correspond to conventions a note appears when the program is started and the date format is set to MM/DD/YYYY.

The year is displayed only by two characters in the status bar.

2.2 General description of the posting process with AIP

In general, posting dialogs are divided into several visual views at AIP. These views (partial dialogs) cover the entire screen and only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. This process is described by way of the following example (interrupt operation). The other dialogs can be operated in the same way.

The “interrupt operation” function is executed. This task is started by clicking the “interrupt operation” function from the second toolbar:

Interrupt operation

The “interrupt operation” dialog opens and the first view is displayed. The function that is currently being executed (in this case: interrupt operation) is shown in the header.

1st view (partial dialog)
The views are consecutive

Posting function that is currently being executed

Interrupt operation S30002090200 Deburring at workplace 33021

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0200 Deburring

Prod.Order S30002090200

Article SAR-VPO011-M-9005

Mirror housing right, bla

Target qty. 1200 PCS

Yield 660 PCS

Scrap 0 PCS

Completion 55%

Quantities that have already been collected (yield, scrap)

General OP data

Yield 22

Scrap 7

Active input field

Virtual keyboard

Cancel Go on

07/12/12 23:08:28

The first view “enter quantities” provides the user with the possibility to enter the produced yield or scrap quantities. The virtual keyboard is shown or hidden automatically, subject to the active input field.

Quantities can be entered using the virtual keyboard or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). Once all values have been entered in the first view, the next view can be opened by clicking the “next” button.

The “cancel” button is displayed in all partial dialogs and allows for the entire posting dialog to be cancelled/closed at any time.

The next view can be opened either by clicking the “next” button or by clicking another tab (in our example: “select status” or “confirm”). Please note in this context, that no view can be skipped when the views are navigated bottom up (view 1 → view 2 → view 3). This means: if you are in the first view (enter quantities) and you click the third view (confirm), the second view (select status) will be displayed first.

Vice versa, when navigating top down (e.g. from the “confirm” view to the “enter quantities” view), every view may directly be opened by clicking at it. In this case, views are actually skipped. But the “back” button also allows for the views to be opened one after the other (top down).

As long as the dialog has not been confirmed, entered data may be changed at any time by scrolling back and forth.

The workplace status that is to be set, once the operation has been interrupted, is determined in the second view “select status”. This status may be chosen from the displayed status list. This list can be restricted using the “filter” field. Once the required values have been entered, the next view can be opened by clicking “next” (in our example it is the last view).

1 Enter quantities

2 Choose status

3 Confirmation

Operation 0010 Assembly

Prod.Order	4199LL12	Target qty.	30000 PCS
Article	34364-647-345	Yield	6833 PCS
	Seating Plate	Scrap	333
		Completion	22%

Workplace 406457A MOUNT1

New status	START UP	Workplace data
------------	----------	----------------

Entered quantities

Yield	22	Quantities posted for the OP.
Scrap	7	

Staff Number

Input field for the badge number

Cancel

Back

Interrupt OP

The partial dialog “confirm” shows a summary of all values entered so far in the dialog. Provided that the user agrees with the entered data, the “interrupt operation” dialog can be confirmed, once the badge number has been entered. Then the dialog including the data is sent to the server and posted.

If an input field of the dialog is not filled out properly (e.g. a mandatory field is empty) the field is highlighted in red in the corresponding view and focused to enable the user to directly correct the field content.



If a workflow dialog is opened it may directly be exited by clicking the ESC button. This is also the case, if the virtual keyboard is opened. Thus, the ESC button cannot be used to close the virtual keyboard.

3 Basic AIP Display

	→	
--	---	--

In general, *AIP* has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons within the touch screen or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these selection lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

3.1 Basic displays – header and footer

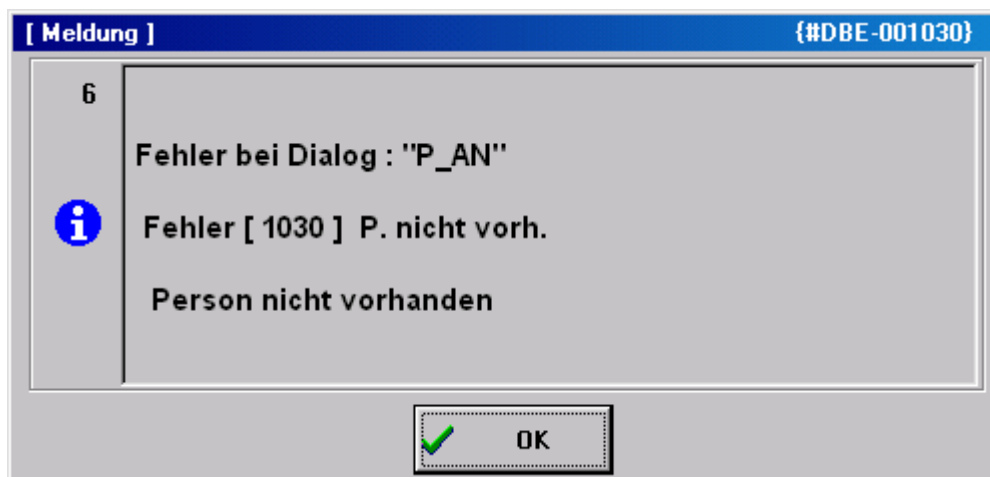
Header



The *AIP* logo is displayed top left of the screen, which may be exchanged by a customer logo after corresponding configuration.

Possible messages (e.g. if a dialog is opened for more than five minutes) are displayed to the right of it.

A separate window opens to display error messages that occur during data collection (e.g. validity checks).



Basic displays

A maximum of 16 workplaces or machines can be assigned to the *AIP* terminal. The individual workplaces can be found within the list area in the order in which they have been assigned to the terminal in MOC.

As regards the basic display of the *AIP* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured within the terminal configuration at the client. The individual basic views are described in the sections that follow.

Footer

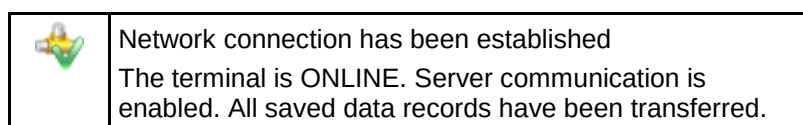


The MPDV logo can be found at the bottom left of the screen. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

AIP status



	Commands are being transferred to the server
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

3.2 Basic display “tabular view“

1st list
Workplaces assigned to the terminal

2nd list
List of registered operations

3rd list (optional)
e.g. list of registered staff

The screenshot shows the AIP (Advanced Input/Processing) interface in 'tabular view'. It consists of three main sections:

- Machines/Workplaces:** A table with columns: Machine/Workplace, Status, Status since, and Note. It lists four entries: 33021 Manual Workplace Deburring, C (PRODUCTION, 28.02.10 06:00), 40312 Manual Workplace Deburring, C (OUT OF MATERIAL, 28.02.10 20:55), 406457A Mounting Place 1 (PRODUCTION, 28.02.10 18:40), and 406457B Mounting Place 2 (PRODUCTION, 28.02.10 15:44).
- Operations on workplace:** A table with columns: Article, Order, Operation, Target qty., Yield, Scrap, N, and T. It shows one entry: SAR-VPO011-M-9005, S3000209, 0200 Deburring, 1200 PCS, 660, 0.
- Registered persons:** A table with columns: Person, Name, First name, and Advance logon. It shows one entry: 10002630, White, Susan.

Below each table are interactive buttons: 'Lock prod.status', 'Modify status', 'Log on operation' for the first table; 'Partial confirmation', 'Interrupt operation', 'Log off operation' for the second; and 'PZE', 'CAQ' for the third. The interface also includes a status bar at the bottom with the date and time: 07/12/12 22:13:38.

The tabular basic display consists of two or three tables, subject to the configurations made. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional display).

“Machines/workplaces” table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as the designation are displayed.

Status

The status is displayed in color and as status text. Coloring is as follows:

- green: Production
- yellow: Assigned status
- red: Not assigned

Status since

Point in time since the status is available.

For ADE workplaces¹ the point in time refers to the last manual status change, for MDE workplaces it refers to the time when the last status change was identified (for machine connections), to the point in time when the status was changed manually the last time or to the time of the last shift change.

Please note

It is indicated here if the “block production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.



Please note

By way of “customizing” services it is possible to adapt the layout of display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic display of terminals. The software does not allow it.

Provided that the “compensate manual quantities” option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately, once they have been entered but only once lists have been reloaded.

¹ We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an ADE workplace.

"Operations at workplace" table

The second table shows the operations that are currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number*.

Target quantity

Target quantity that is defined for the operation.

Yield

Yield which has already been produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity which has already been produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note, which should be visible at the terminal, has been recorded for this operation in the graphic planning board at the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a line that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured respectively. Which information is displayed in this list depends, among other things, on the workplace configuration.

The following lists can be displayed:

- Staff logged on to the currently selected workplace (BDE)
- Resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow for switching between these lists.

Please note

The registered staff displayed in the third list correspond to the list of the dialog “F5 registered persons...”. Selecting a person in the third list does *not* affect selection of the operation in the list of OPs running at the workplace and, as a result, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic display

A toolbar, which may be configured by customizing, is assigned to each list in the basic display. This makes the purpose of the function clear to the user. The “partial upload/confirmation” function can be found, for example, below the list of registered operations.

In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. change partitioning) can be opened by clicking the corresponding button.



Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

3.3 "Machine overview" basic display

If the “change view” button is clicked in the basic display, the view changes to the following presentation:

Toolbar of assigned machines

60610	60611	60612	60614	60623
-------	-------	-------	-------	-------

Workplace/Machine	60611	Status	TOOL FAULT	
Short description	ARB-320S	Start	14:00	10.11.2010
Group	SGM2	Duration [hh:min]	15368:30	
Divisor	1	Yield	1018 PCS	
Target cycle [sec/cycle]	21	Scrap		
Actual cycle [sec/clock]	0	Cycles		
		Note		

Machine information

Operation	S61003280100	Target qty.	95000	PCS
Article	SAL-OSI011-K-9001	Yield	1018	PCS
Article Designation	Mirror housing left, crem	Scrap	23	PCS
Remark 1		Target qty. since logon	---	PCS
Remark 2		Yield since logon	0	PCS
Plan. duration	554:10	Deviation [%]	---	
		Completion	1%	

Order information

07/12/12 22:30:54

This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation altogether has three areas:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines as well as to shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. Provided that several orders/OPs are logged on, it can be switched between them using arrow keys that are then shown additionally.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$\text{Displayed partitioning} = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \dots \right]$$

TLG _M	Partitioning of the machine (TLG in mnr.lst)
DIV _M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG _{AGi}	Partitioning of the individual order (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview at the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal may get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) at the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time in which the production lock of the machine has not been active). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:

$\text{TargetSinceLogon} = \text{NetRunningTime}[\text{sec}] * \text{Partitioning} / \text{TargetCycle}[\text{sec/stroke}]$

NetRunningTime: Time since logon, in which the production lock has not been set. This calculation does not take into account any breaks that might be defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: $\text{Deviation}[\%] = 100\% * (\text{YieldSinceLogon} - \text{TargetSinceLogon}) / \text{TargetSinceLogon}$

Completion

The bar represents the proportion of "yield", which has been produced until now, compared to the "target quantity".

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced by a picture showing a yellow or red oilcan, depending on the maintenance activity that is required:



3.4 "Machines as icons" basic display

This view can be configured as the default display within the terminal configuration at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the "Production" status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon presentation of machines”.

4 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
-----	-----	---- General ---
00.	00.ABC123	Data not defined, → 00. will be deleted and data "ABC123" will be passed to standard processing
01.	01.OK 01.ESC	Action barcode → Dialog cancelled or ended with OK button or Esc button.
-----	-----	---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Suborder number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16.200	Machine → Acronym <MNR> → Passed to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> → Passed to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> → Passed to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > → Passed to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> → Passed to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
-----	-----	----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---
40.	40.100 40.MONTAGE	Destination → Acronym <ZLO> → Passed to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> → Passed to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→→
43.		Throughput batch number → Acronym <DLL>→→
44.		Alternative batch number → Acronym <CNR:ALT1>→→
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
-----	-----	---- Mainly for the HYDRA-PZE module ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> → Passed to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> → Passed to dialog with FGR=1 or FGR=1001
-----	-----	---- Customer-specific barcodes ---
90.		
...		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]

BarcodePrefixChar=\$



If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font "Codedreineun" and prefix "."

Prefix	Barcode	Raw data
10.	*10.123456780100*	ANR = 123456780100
	10._ABCD12340100	ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

4.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR BarKenn91=EGR:GUT	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345" → SAPCNR=12345

Firmly assigned barcode prefixes:

10:ANR
 11:AUNR
 12:AFOLG
 13:AGNR
 14:UAGNR
 15:RMNR
 16:MNR

 17:MST
 18:AUSGRD
 19:AGGGRD
 20:BPOS
 21:LPKZ
 22:SPLNR
 40:ZLO
 41:TPE
 42:CNR
 43:DLL
 44:CNR:ALT1
 45:CNR:ALT2
 46:CNR:ALT3
 47:CNR:ALT4
 48:CNR:ALT5
 49:CNR:ALT6
 50:KNR
 51:PNR
 52:KST
 53:FGR

5 Local Configuration File ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the c:\ctaip directory.



Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

5.1 Basic configuration

Entry	Comment
Section [system]	
Usr=21	Distinct terminal number
Hypath=d:\hydra\ Hypath=/usr/hydra/	HYDRA server path: Windows NT: -> DOS notation, the drive is the local drive of the server on which HYDRA or xMES is installed Unix: -> Unix notation
Hostname=192.9.200.24	Internet address of the server
Offlinetimeout=600	In offline mode, the interval after which online access should be attempted the next time. The interval is specified in seconds
Showcursor=on	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
Loadfile=ctnet\win\ctaip.txt	Configuration file for downloading the application from the server. The path is relative to the server directory (i.e. within "hypath")
Watchdog=on	ON: Watchdog is activated OFF: Watchdog is not activated
Demo=off	'on': Offline demo mode; always off in the production environment
parameters=-t	The -t parameter switches off the virtual keyboard
TMOUT_C=xxx	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_S=xxx	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds

Entry	Comment
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	Configuration of customized barcode prefixes.
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
Com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to the connected devices MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
ZAEHLER= 1 2 3 4 5 6 7 8	Assignment of physical inputs of the MSS (machine interface) to logical counters (ZAEHLER) according to configuration: The first connector (labeled “0” on the MSS) corresponds to the logical counter no. 1 Please note: MSS1 has only 8 inputs If digital inputs are also to be used with MSS1, configuration should be changed as follows: ZAEHLER= 1 2 3 4 IN= 5 6 7 8
IN= 9 10 11 12 13 14 15	Assignment of physical MSS inputs to logical inputs as per configuration: The ninth connector (labeled “8” on the MSS) corresponds to the logical input no. 1
CHARGE= 5 6	For batch recording: Inputs for automatic batch changes In this case, the connectors 5 and 6 are the inputs for the automatic batch change.
MSSZyklusBerechnung=ON MSSZyklusReferenz=0	Activates reading out of cycle time values from the machine interface (MSS). Reference for cycle calculation (smallest time unit of the MSS). The default value is 0, if the parameter is not specified. 0 corresponds to 100 ms. 2 corresponds to 20 ms.
CalculateCycle=ON	The terminal itself calculates the actual cycle. If the connected control does not provide a determined actual cycle, then a calculated actual cycle can be displayed: This calculated value is also available for DS100 terminals. Please note: The calculation cannot provide exact values.
WochenEnde_ProdCheck=ON	This function prevents the weekend automatic from affecting the “production” status and the workplace from being set to status 999. ON is set by default In case WochenEnde_ProdCheck=OFF, the automatism switches to status 999.

Entry	Comment
sFrom999ToNotAttributed=OFF	Only affects HYDRA-MDE machines with operation mode = "no monitoring". If the weekend automatic function is enabled this option prevents the machine from switching to the "not assigned" status when the weekend automatic ends. Reasons: The "not assigned" status may not be set manually for machines that are configured with the "no monitoring" option. The "not assigned" status is normally only set for machines with operation mode = "cyclic monitoring" or "monitoring by operating signal".
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be called to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [PDV]	
Modus=PDV Modus=PDV,BDE	Operation as interface IOP↔DOS terminal (standard) Operation as shop floor terminal with PDV
Terminals=121,122,123	CTDOS terminals "connected" by LAN. Only required with Mode=PDV
PDVTerminalDir=c:\hsrv\spool\ PDVIOPTDir=c:\IOPSim\	Directory for communication with CTDOS terminals Directory for communication with the IOP
For debug purposes only: InfoFenster=100 SlowDown=600	Number of lines of the "current" window (presentation of last PDV actions) Slow down, to make events "visible"
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.

Entry	Comment
BarcodeNest=	This field specifies which acronym is entered into the cavity number field by the scanner.
BarcodeNumm=	This field specifies which acronym is entered into the number field by the scanner.

6 Central Configuration File hytnrcfg.ini

This file includes different configurations for all or single terminals at a central place.

Each section is available in a generally accepted version

[section 0].

However, entries included in this section can be overwritten by entries in a terminal-specific section [section <TNR-USER>]

<TNR-USER> = HydraUser = Terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/HYDRA User



The hytnrcfg.ini file is loaded from the server every time the terminal is started.

Section / Entry	Comment
	→→→
[Tnr configuration 0]	
FollowExternStatus=on	Transfer of machine statuses when reloading machine list Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
InstallFonts=on	If this is set to "off" fonts will not be installed during the restart. ON=DEFAULT
OnlyInstallFontsAfterDownload=false	"InstallFonts=on": If true then fonts will only be installed directly after a download. If false then fonts will be installed every time the terminal is restarted. (false = DEFAULT)
InstallTvicport=on	If "off" the LPT driver "tvicport.sys" will not be installed. It is required for HYDRA-ZKS. ON = DEFAULT
[Terminal->USR 0]	

AttachedApplication=First	This configuration checks whether or not an application is connected in Windows that matches the file extension of the document to be displayed from the OP info dialog. If there is such an application, it will be used for displaying the document. If there is no connection, viewers configured in ctaip.ini (→ [ext. software]) and internal viewers will be used. In case, an extension is completely unknown it is attempted to display it as text Different settings may be configured: First → search for connected application first AfterUserViewer → If a UserViewer is configured this one overrides the connected application (also applies for ExcelViewer, WordViewer and PowerpointViewer) Last → Only if no ctaip.ini assignment is found for the file extension, then the connected assignment will be searched for (default). Off → Connected application is never searched.
HTTPBrowser=standard	Viewing of documents (via OP info): If documents are configured with a path of the <u>type</u> "http", the file will not be downloaded to the terminal, but the link will only be transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows will be used.
SupressErrorMessage=70012	Suppress message "material is not planned"
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field "user" can be edited in the terminal (by default: no editing) true → activates keyboard input for the "user" field in the terminal

Transparency=255	The signature dialog can also be transparent. 255 → Signature dialog is 0 % transparent (not transparent) 1 → Signature dialog is 99% transparent (maximum transparency) (Default = 155) Available as of CTAIP (V# 2.0.2.25) / CTWIN (V# 7.2.5.99)
ShowPosition=TR	The position of the signature dialog can be adjusted as follows: TL Top – Left TM Top – Middle TR Top – Right ML Middle – Left MM Middle – Middle (Default) MR Middle – Right BL Bottom – Left BM Bottom – Middle BR Bottom – Right Available as of CTAIP (V# 2.0.2.25)
USE_SERVICE_ACCOUNT=1	0 (default) SSO: do not use ServiceAccount (requires the terminal to be started with the "user" domain (SSO)). Please note: ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY	REPORTING_USER_READONLY The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field is read-only. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. REPORTING_USER_CHANGEABLE The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field can be modified. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible.

SIGNATURE_1_LOGON_TYPE=HYDRA	"" / Not set / "EMPTY" There is also an alternative login procedure. HYDRA The tab identifying users via the Windows user is blocked. The HYDRA user must be used for identification purposes. This requires, however, that in the HYDRA HR master all users logging in are created and that the "SSO" option is not set. Otherwise, successful authentication is impossible. ACTIVEDIRECTORY The tab identifying users via the HYDRA user is blocked. The Windows user must be used for identification purposes. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. MIXED_BUT_UNIQUE Either the HYDRA or Windows login procedure is available, subject to whether or not the "SSO" option is set for the registered user in the HYDRA HR master. "SSO" enabled → Windows only "SSO" disabled → HYDRA only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures with the terminal in the area of quality data collection.

6.1 Layout configuration

Entry		Comment
Section and/or	[terminal configuration 0] [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE		In case of an error in reading the clock (e.g. after coming out of standby mode), this configuration makes sure that the time is accepted without having to confirm a dialog. Afterwards the terminal time will be synchronized with the server time using a PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON		As of ctaip V# 2.0.2.23 Prevents the warning after restarting the terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default = OFF)

Entry	Comment
NetRuntimeMode=2	As of ctaip V# 2.0.2.50: Alternative calculation of the target quantity since logon: The net run time is not calculated from the times when the production lock is enabled (PSperre=green) but only from the shift times less the shift breaks. Consequently, it can also be displayed, even if the terminal program has been restarted.
Section [QRD-PRINTER->TICKET 0] ;(general configuration) [QRD-PRINTER->TICKET 2xxx] ;(2XXX configuration for a specific terminal)	
COMPLETE-ABSENCE-OF-LOCAL-MNR- DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured <Events>, if it is not available locally => This configuration might be required/necessary for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR- DATA-FOR-EVENT=< Events >	Reloads the order row for the configured <Events>, if it is not available locally ➔ This option has been implemented to access order data within the master data, e.g. when logging an order on.
COMPLETE-...-EVENT=< Events > COMPLETE-...-EVENT=#ALL# COMPLETE-...-EVENT=A_AN A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN A_P_AN> restricts reloading of information to the specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.

7 Collection of Serial Numbers

General

Serial numbers are assigned to be able to differentiate between individual items of material. [Wikipedia](#) provides the following definition for serial numbers:

A **serial number** (also manufacturer's serial number or MSN) is a unique, alphanumeric code assigned for identification of a single unit. Although usually called a number, it may include letters, though ending with digits. Serial numbers are used to identify elements pertaining to series production also providing information about production conditions and thus enabling traceability of the used components, also for electronic products.

Serial numbers can be:

- generated/assigned at the end of the production process
- generated/assigned in earlier stages of the production process and monitored by the process
- integrated as used components in the production process and monitored by the process



[Merging serial numbers](#) represents the assembly of several components listed by serial number into one component part listed by serial number.

MES supports manufacturing businesses in many ways with the collection of serial numbers and enables complete traceability based on the data recorded in MES.

Definition of the supported variants

MES provides different variants to record serial numbers:

Variant	Description of the variant	Description	Documentation
---------	----------------------------	-------------	---------------

Variant	Description of the variant	Description	Documentation
1	Entry of serial numbers for OPs that are not subject to management in batches	Recording of serial numbers for operations that are not managed in batches enables the collection in relation to serial numbers if incorporated materials their batches and consumptions are not required to be recorded. The production order specifies the serial numbers to be processed as part of a production order. For each operation the user identifies the serial numbers and classifies them according to the produced quality. The system makes sure that this process can only be performed once for each serial number and operation.	here
2a	Entry of serial numbers for OPs that are subject to management in batches - manual input of the serial number	The user enters the serial number and the system is informed about it. In this case, the serial number is generated in another system. The recorded serial number is assigned a HYDRA batch number.	here
2b	Entry of serial numbers for OPs that are subject to management in batches - automatic assignment of serial numbers	In this case, the system generates the serial number in an unambiguous format applicable to the whole system. Serial numbers are generated according to the rules applying for the generation of batch numbers. In this case, the serial number matches the HYDRA batch number.	here

Variant	Description of the variant	Description	Documentation
2c	Entry of serial numbers for OPs that are subject to management in batches - automatic assignment of serial numbers	In this case, the system generates the serial number in an unambiguous format applicable to the whole system. Serial numbers are generated according to the rules applying for the generation of batch numbers. The serial number is assigned a HYDRA batch number.	Here
2d	Entry of serial numbers for OPs that are subject to management in batches - automatic transfer of serial numbers by using input batches.	In addition, to the conventional generation of output batches, the system particularly supports the generation of merged batches including individually assigned serial numbers. There is a direct connection between individual serial numbers and the generated output batch (merged batch), which logically embraces the serial numbers. Serial numbers are taken over as individual batches provided by logging an input batch/merged batch on. The serial number is the batch number. In addition, individual serial numbers are assigned to a superordinate merged batch.	This document

HYDRA batch number vs. serial number

In HYDRA the [HYDRA batch number](#) is unambiguous throughout the entire system. The serial number is kept throughout the complete process, provided that [serial numbers have not been merged](#). Consequently, it is differentiated between the internal and external HYDRA batch number to represent the collection of serial numbers:

HYDRA batch number

The HYDRA batch number is the ID by which the serial number is managed in HYDRA throughout the complete collection process.

The HYDRA batch number is used on labels to identify the material.

To ensure unambiguity within the system, an internal HYDRA batch number is assigned and managed for every HYDRA batch number when recording serial numbers.

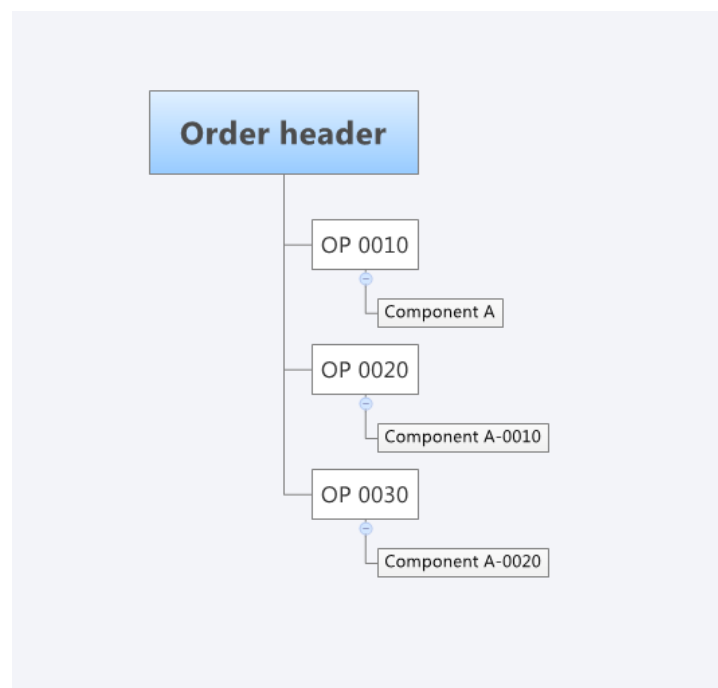
Internal HYDRA batch number

The internal HYDRA batch number represents a system-wide, unambiguous ID identifying batches and serial numbers as an alternative to the HYDRA batch number.

In contrast to the HYDRA batch number; this ID has to be changed from production step to production step to keep unambiguity.

This example explains the procedure:

A production order has the following structure:



The interface provided a merged batch with the serial number "BXC6GF7H". Upon posting this serial number, the HYDRA batch number and the internal HYDRA batch number change as follows:

Process step	Batch status	HYDRA batch number	Internal HYDRA batch number	Comment
Status <u>after</u> transferring the number to MES via the material interface and <u>prior</u> to logging the input batch on.	FREE	BXC6GF7H	BXC6GF7H	Initial status when the HYDRA batch number and the internal HYDRA batch number are identical.
Status after logging the merged batch on to OP 0010	RUNNING	BXC6GF7H	BXC6GF7H	No changes
Status after entering the serial number at OP 0010	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	FREE	BXC6GF7H	PR1111X112	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number
Status after entering the serial number for the OP 0020	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X112	The "old" batch is assigned the "processed" status.

Process step	Batch status	HYDRA batch number	Internal HYDRA batch number	Comment
	Free	BXC6GF7H	PR1111X113	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number
Status after entering the serial number for the OP 0030	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X112	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X113	The "old" batch is assigned the "processed" status.
	Free	BXC6GF7H	PR1111X114	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number

As regards a HYDRA batch number (serial number), there might exist 1-n HYDRA batches at different times, whereas only one of the batches may be "free" or "running" at a time.

Please note for the supply via interface

Merged batches and serial numbers can be transferred from other systems (e.g. ERP systems) to MES via the [material staging interface](#). Some particularities have to be taken into account when transferring the merged batch/combination of serial numbers:

- This order has to be observed when transferring serial numbers and their merged batches from Navision to HYDRA:
 1. Transfer of serial numbers
 2. Transfer of the merged batch record
- The below table shows how specific fields are assigned differently when being transferred to HYDRA.

Logical field	Interface field	Merged batch	Serial number
HYDRA internal batch number	HY_LOSNR	Merged batch number specified by ERP	Serial number specified by ERP
HYDRA batch number	DLL	Merged batch number specified by ERP	Serial number specified by ERP
PPS batch	PPS batch	Batch number specified by ERP	Batch number specified by ERP
Merged batch number	MCNR	BLANK	Merged batch number specified by ERP
Merged batch ID	SLOS	Fixed "J" (upper case "j")	BLANK
Merged batch type	SLOSTYP	Fixed "J" (upper case "j")	BLANK

Logical field	Interface field	Merged batch	Serial number
HU level	HULEVEL	1	0
Inventory type	BESTART	Fixed "H"	"S"
Number of individual batches	SLOS	Number of included serial numbers	0

Posting of serial numbers

If a merged batch with assigned serial numbers is finished, the system updates data and generates new documents in the form of log records and goods movements.

Transfer of data from the operation

This data is taken over from the running operation and transferred to the merged batch and the assigned serial numbers:

- Material and material name
- Material type

Transfer of entered data to serial numbers and the merged batch

The machine, producing operation, person performing the posting, material buffer and the transport unit are transferred to the merged batch and the serial numbers included in the merged batch.

Any comment that might be entered is transferred to the merged batch as additional information on the batch.

The status of the merged batch and serial numbers is set subject to the specified quality/class of the output batch:

Selected class	Assigned batch status	Reason accepted
Yield	Free	No
Scrap	Locked	Yes
Rework	Free	Yes

Selected class	Assigned batch status	Reason accepted
Open quantity	Free	Yes

Batch attributes collected in relation to the merged batch are transferred to the merged batch as batch attributes.

Batch attributes collected in relation to the serial number are transferred to serial numbers as batch attributes.

Document links collected in relation to the merged batch are transferred to the merged batch as document links.

Document links collected in relation to the serial number are transferred to serial numbers as document links.

Quantities of the merged batch

The generated merged batch includes the total quantities of the serial numbers assigned to the merged batch.

The number of assigned serial numbers is defined for the merged batch as number of individual batches.

Activities for serial numbers

New batches with a new, internal HYDRA batch number are generated for the serial numbers included in the merged batch (for further details on this procedure, please see section HYDRA batch number vs. serial number) and the preceding batches are assigned the "processed" status.

Generation of ADE log records

An H record is generated for the merged batch. This one includes the quantity(ies) of the merged batch, which in turn result(s) from the quantities of the assigned serial numbers. The batch number of the merged batch is stored at the H record.

Separate log records are not generated for serial numbers.

Generation of goods movements

The generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches.

Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.

Details on how to configure goods movements can be found in the [configuration guidelines for serial number recording](#).

Tracing

Tracing information is updated for the merged batch and serial numbers in the system:

- The generated merged batch is connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.
- The serial numbers included in the merged batch are also connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.

[Establishing the connection](#) between serial numbers and input batches can be disabled if required.

8 Collection of Serial Numbers at AIP

Usage

You use the "enter serial number" dialog to enter and track serial numbers in the production process.

Prerequisite

Various configurations have to be made in the system to use the dialog. Further details can be found in the document dealing with the [Description of the configuration to use collection of serial numbers](#).

"Collection of serial numbers" dialog

The "enter serial number" dialog (A_SNR_A) consists of workflows providing extensive possibilities for data collection. By default, the dialog includes the following workflow steps:

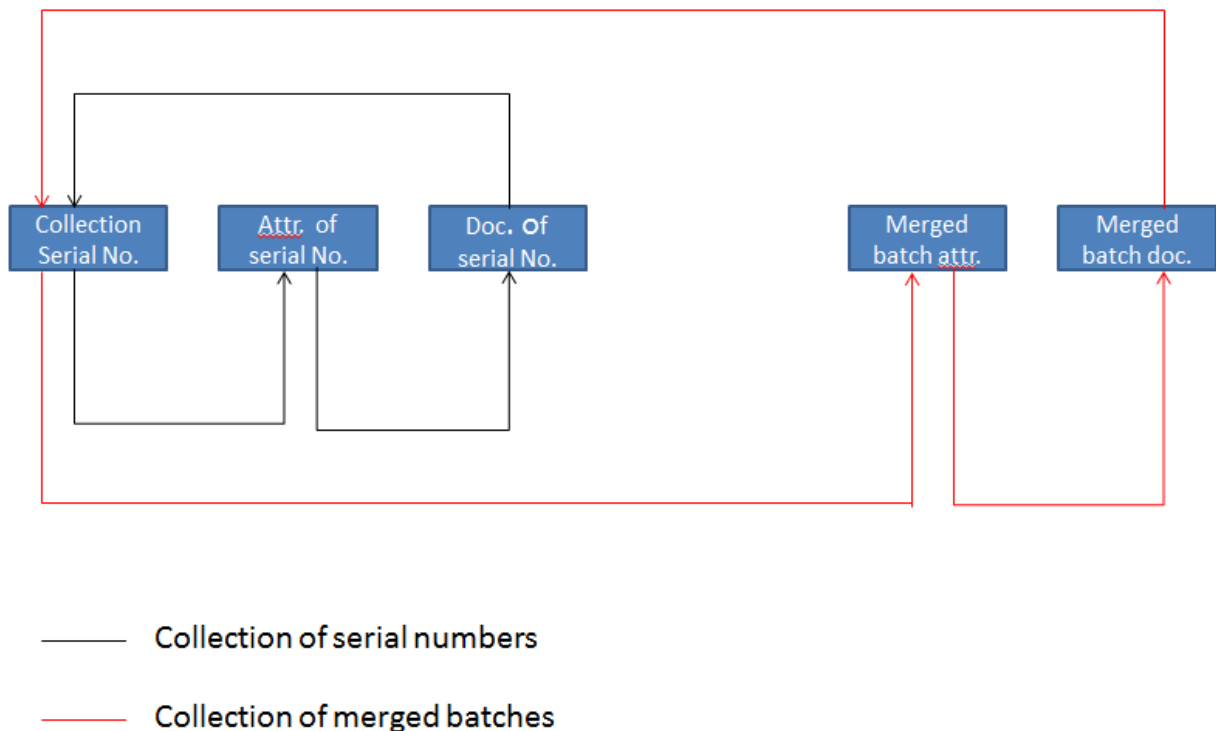
Workflow step/detailed dialog	Technical name	Usage
Collection of serial numbers	WF_A_SNR_A	This workflow step shows basic data of the serial number or the user can enter this data here.
Merged batch attributes	WF_SNR_ATTR	Entry of batch attributes for the merged batch
Merged batch documents	WF_SNR_DOC	Entry of document links for the merged batch
Serial number data	SNR-DATA	Detailed dialog to enter additional data specific to serial numbers. The detailed dialog can only be started for a single serial number.

Depending on the active workflow step, different function keys are provided.

Basically, data is collected in two steps:

- Data collection for serial numbers
Data is collected for serial numbers until a new merged batch can be completed.
Data collection restarts, once a merged batch has been completed.
- Data collection for merged batches
Basically, data is collected for the merged batch when the merged batch is completed; but it may also be started at an earlier point in time.

The following diagram illustrates the two acquisition cycles:



Workflow step: "collection of serial numbers"

The workflow step "collection of serial numbers" has been designed to enter data regarding one or several serial numbers.

The dialog shows the following data that cannot be modified:

Workplace

Current workplace as selected in the basic screen of the terminal.

Operation

Currently running operation as selected in the basic screen of the terminal.

Material

Material produced by the operation as laid down in the operation's article/item.

Merged batch

Merged batch number as defined by the system for the next merged batch to be completed.

The user may enter the following data:

Serial number

Subject to the type of serial number collection, the "serial number" field can be used to:

- enter serial numbers manually, instead of selecting them manually from the table of serial numbers
- scan serial numbers, instead of selecting them manually from the table of serial numbers
- If serial numbers have to be entered manually, they can be entered or scanned manually.

Staff badge number

Entry of the staff badge number is mandatory.

Quality

The user may define the quality. A reason has to be entered for the ratings "scrap", "open" and "rework".

Material buffer (*1)

The material buffer defined for the machine is suggested as the target material buffer.

Comment (*1)

A comment may be entered for the merged batch.

Transport unit (*1)

Suggests the transport unit that is identified as the default unit in the [assignment of material types to transport units](#) regarding the material type of the operation.

If nothing is assigned, "SYSTEM" is suggested.

Transfer (for each merged batch / all merged batches from now on)

For the fields labeled with (*1) it may be specified if they only apply for the merged batch that is currently being processed or for this and all future merged batches of this operation.

This definition applies until the operation is interrupted and/or logged off, even for different shifts.

The table of serial numbers shows the serial numbers of the registered input batch that have not yet been classified for this operation. The quality status (Q status) of serial numbers may be filtered, provided this status is set by quality data collection.

Single serial numbers can be selected by double clicking the table. Selected serial numbers are highlighted at the left-hand side of the table by an "X" with green background. There are also function keys for mass selections.

There is a number of function keys facilitating dialog handling, to input data for serial numbers or to complete a posting:

Designation	Usage
Close	Closes the dialog.

Designation	Usage
Complete	The currently generated merged batch is completed for the entered/assessed serial numbers. The merged batch number is displayed on the terminal. If the user confirms it, the dialog can be continued and remains opened to enter additional merged batches.
Select all	All serial numbers displayed in the list of serial numbers are selected.
Unselect all	All serial numbers displayed in the list of serial numbers are deselected.
Invert selection	Subject to the current state of selection, the system changes it: • Selected serial numbers are unselected • Unselected serial numbers are selected
Serial number data	Detailed dialog to enter data specific to serial numbers. The detailed dialog can only be started for a single serial number.
Next	Leads to the next workflow step

Detailed dialog "serial number data - batch attributes"

Batch attributes may be entered for the serial number in the "serial number attributes" workflow step of the "serial number data" detailed dialog.

The "attributes" function key allows entering attributes for a batch or for all serial numbers. The input dialog suggests all attributes that are assigned the "Capture attribute while generating batch" option in the [configuration of batch attributes](#) regarding the operation's material type.

The "go on" function key allows entering document links for serial numbers.



The list only shows the recorded batch attributes; batch attributes that might already be available are not shown.

Workflow step - merged batch attributes

Batch attributes may be entered for the serial number in the "merged batch attributes" workflow step.

The "attributes" function key allows entering attributes for the merged batch that is currently being processed or for this and all future merged batches of this operation. The input dialog suggests all attributes that are assigned the "Capture attribute while generating batch" option in the [configuration of batch attributes](#) regarding the operation's material type.

The "go on" function key allows entering document links for the serial number.



The list only shows the recorded document links; document links that might already be available are not shown.

9 Configuration for the Collection of Serial Numbers

Activation at machine/workplace

These configurations have to be made for the [machine / workplace](#) to enable the collection of serial numbers:

Parameter name	Value
Workplace configuration → Workplace master data → Workplace category	N Machine
Workplace configuration → Workplace master data → Workplace type	E Single workplace
Workplace configuration → MPL → Batch management	L Batch tracing (input/output batches)
Workplace configuration → MPL → Preceding material buffer	If required, define a preceding material buffer .
Workplace configuration → MPL → Subsequent material buffer	If required, define a subsequent material buffer .
Workplace configuration → MPL → Automat. generation of batch numbers	J Automatic generation of batch numbers for production batches (MPL) enabled

Maintain material types - for the operation

Maintain the [material types](#) to be defined for the operation and adapt them to your specific requirements of data collection.

Maintain material types – for components

Maintain the [material types](#) to be defined in the component list and adapt them to your specific requirements of data collection. Configure at least the following values for the component that is integrated as merged batch including assigned serial numbers in the operation:

Parameter name	Value

Parameter name	Value
Input batch processing → Inventory management	R - Yes, backflush (retrograde)
General → Options → transferred at interface	Enable this option if you expect consumption postings or final backflushes (notification of goods receipts) for this material type. Then another configuration can be used to control the transfer of merged batches and serial numbers separately.

Maintain reasons

Maintain [reasons for scrap and rework as well as reasons for open quantities](#), if you use these quality classes.

Maintain transport units

Create [transport units](#) in the system if you want to use them for data collection. Assign transport unit ans according material type.

Assign material types to transport units

If you want to use [transport units](#) and you created them in the system, you may assign them to the [material types](#) maintained in the system and define a default transport unit for each material type. This one will then be selected in advance in the input dialog.

Perform the assignment in the [assignment of TPU to material type](#).

Activation at the operation

These options have to be set for the operation.

Parameter name	Value
Batch management requirement	Yes
Serial number requirement	A Collection of serial numbers

These options are available to edit this information:

Manual maintenance in the operation

Edit the options manually for the [operation](#)

Maintenance of processing code (customizing)

Edit the options for the [processing code](#).

Maintenance of the template (customizing)

Edit the options of the [processing code](#) with value "A" - collection of serial numbers and assign the processing code to an [operation template](#).

Explicit specifications for the interface (recommended procedure)

Explicitly transfer the options for the operation at the interface.

Activation in components list

Maintain the input quantity in the [component list](#) by entering "1.0" units for components listed as merged batch including assigned serials numbers. Assign the relevant material types to the components.

Activation of the link between serial number and input batch

For tracing it is necessary to link the incorporated input batches with the produced serial numbers. To enable this processing, a configuration has to be enabled in [INI configuration](#). The entry itself is part of the default delivery and only needs to be enabled.

Parameter name	Value
INI name	MPL
Section	SERIALNUMBER
Key	CONNECT_SNR_CNR
Value	Y
Active	Enable the entry

Configure batch attributes for data collection when recording serial numbers

If you want to enter batch attributes manually when collecting serial numbers, create the [batch attributes](#) to be recorded in the system in relation to the [material type](#) of the operation. To do so, maintain at least these configurations:

Parameter name	Value
Options → Capture attribute while generating batch	Enable the option.
Options → Position	Specify the position - the system sorts the configured attributes in an ascending numeric order (bottom up).
Data type	Maintain the data type and length of the attribute to be recorded

Control the generation of goods movements

Define for incorporated merged batches and serials numbers as well as for produced merged batches and serial numbers if you require goods movements subject to uploads to be provided. To do so, configure the following settings in [advanced object configuration](#):



The goods movement option has to be enabled for the relevant material type to be able to use this configuration.

Parameter name	Value
Configuration for goods issues (consumptions)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_A
Parameter	CREATE_MOVEMENT
Parameter value	Y

Parameter name	Value
<i>Configuration for goods receipts (generated material)</i>	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_E
Parameter	CREATE_MOVEMENT
Parameter value	Y

10 Merging Serial Numbers

Usage

Merging serial numbers allows for different material components listed by serial numbers to be combined specifically into one new component part. The serial number of the new component part may be the number of one of the incorporated serial numbers or assigned anew.

Process of merging serial numbers

Merging serial numbers is based on the management of serial numbers incorporated in [merged batches](#) registered as input batches. Incorporated serial numbers are assigned based on the serial numbers included in these merged batches.

This process also outputs a serial number. There are two strategies to get the serial number:

- An integrated serial number is continued keeping the ID
The indicator "superordinate serial number" at the component specifies which one of the incorporated components includes the serial number ID that is to be continued. These components and/or their serial numbers are designated as superordinate serial numbers.

For each operation only one component can be assigned the flag "superordinate serial number".

- A new serial number is assigned
If no component is identified as superordinate, the new serial number to be assigned can either be specified or the system assigns a number automatically.

If a merged batch with assigned serial numbers is finished, the system updates data and generates new documents in the form of log records and goods movements.

Posting of serial numbers

Transfer of data from the operation

This data is taken over from the running operation and transferred to the merged batch and the assigned serial numbers:

- Material and material name
- Material type

Transfer of entered data to serial numbers and the merged batch

The machine, producing operation, person performing the posting, material buffer and the transport unit are transferred to the merged batch and the serial numbers included in the merged batch.

Any comment that might be entered is transferred to the merged batch as additional information on the batch.

The status of the merged batch and serial numbers is set subject to the specified quality/class of the output batch:

Selected class	Assigned batch status	Reason accepted
Yield	Free	No
Scrap	Locked	Yes
Rework	Free	Yes
Open quantity	Free	Yes

Batch attributes collected in relation to the merged batch are transferred to the merged batch as batch attributes.

Batch attributes collected in relation to the serial number are transferred to serial numbers as batch attributes.

Document links collected in relation to the merged batch are transferred to the merged batch as document links.

Document links collected in relation to the serial number are transferred to serial numbers as document links.

Quantities of the merged batch

The generated merged batch includes the total quantities of the serial numbers assigned to the merged batch.

The number of assigned serial numbers is defined for the merged batch as number of individual batches.

Activities for serial numbers

New batches with a new, internal HYDRA batch number are generated for the serial numbers included in the merged batch (for further details on this procedure, please see section [HYDRA batch number vs. serial number](#)) and the preceding batches are assigned the "processed" status.

Generation of ADE log records

An H record is generated for the merged batch. This one includes the quantity(ies) of the merged batch, which in turn result(s) from the quantities of the assigned serial numbers. The batch number of the merged batch is stored at the H record.

Separate log records are not generated for serial numbers.

Generation of goods movements

The generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches.

Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.

Details on how to configure goods movements can be found in [configuration instructions for the collection of serial numbers](#).

Tracing

Tracing information is updated for the merged batch and serial numbers in the system:

- The generated merged batch is connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.
- The serial numbers included in the merged batch are also connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.

[Establishing connections](#) between serial numbers and the input batch can be disabled if required.

11 Merging Serial Numbers on the AIP

Purpose

You use the dialog "Merge serial numbers" to group several materials that are listed by serial numbers. These materials are then listed under one number and can be used and planned more easily in the system. The system saves all serial numbers that are merged in one number.

Requirements

You must make different configurations in the system to use the dialog. For further details, refer to the document [Description of the configuration to use merged serial numbers](#).

Dialog *Merge serial numbers*

The "merge serial number" dialog (A_SNR_U) is a workflow dialog, which provides extensive possibilities for data collection. By default, the dialog includes the following workflow steps:

Workflow step / dialog	Technical name	Purpose
Merge serial numbers	WF_A_SNR_A	This workflow step shows basic data of the serial number or the user can enter this data here.
Merged batch attributes	WF_SNR_ATTR	Recording of batch attributes for the merged batch
Merged batch documents	WF_SNR_DOC	Recording of document links for the merged batch
Merge	A_SNR_MARRIED	Dialog to identify and assign the different ingoing serial numbers
Serial number data	SNR_DATA_MARRIED	Dialog to enter additional data specific to serial numbers. The dialog can only be started for a single serial number.

Depending on the active workflow step, different function keys are provided.

Basically, data is collected in two steps:

- Merge serial numbers

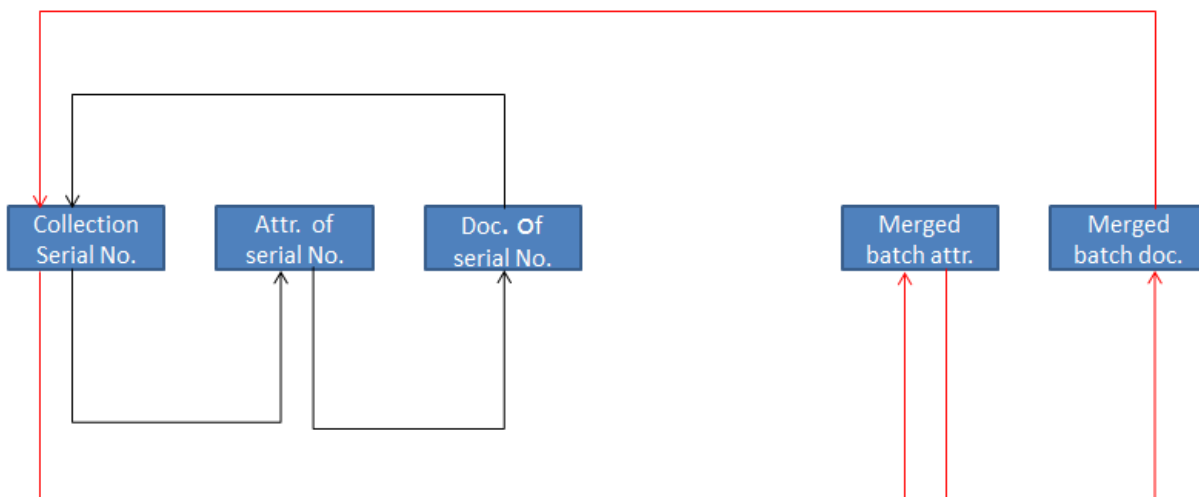
In the first step two or more serial numbers are merged. This process is completed and the "merge serial number" workflow step is opened again showing the superordinate serial number in the table.

Now either additional serial numbers can be added and merged or further data (batch attributes and document links) is recorded for an already merged serial number.

- Collection for merged batches

Basically, data is collected for the merged batch when the merged batch is completed; but data collection can also be started at an earlier point in time.

The following diagram illustrates the two cycles of data collection:



— Collection of serial numbers

— Collection of merged batches

Workflow step: merge serial number

You use the workflow step "Collection of serial numbers" to enter data for one or several serial numbers.

The dialog shows the following data that cannot be modified:

Workplace

Current workplace that is selected in the main view on the terminal.

Operation

Currently running operation that is selected in the main view on the terminal.

Material

Material produced by the operation that is specified for the article of the operation.

Merged batch

Number that the system assigns to the next merged batch that is set up.

The user can enter the following data:

Staff badge number

The input of the staff badge number is mandatory.

Quality

The user can specify the quality. You must enter a reason for the qualities "scrap", "open" and "rework".

Material buffer (*1)

The material buffer defined for the machine is suggested as the target material buffer.

Comment (*1)

You can enter a comment on the merged batch.

Transport unit (*1)

The system preassigns the transport unit that is identified as default unit for the material type of the operation in the [Assignment material type - transport unit](#).

If no assignment is available, "SYSTEM" is entered.

Transfer of data (for each merged batch / all merged batches from now on)

For the fields labeled with (*1), you can specify if the field specifications only apply for the merged batch that is currently being processed or for this and all future merged batches of this operation.

This definition applies until the operation is interrupted and/or logged off, also if the shift changes.

The serial number table shows the superordinate serial numbers that have already been merged and, therefore, are part of the currently running merged batch.

Single serial numbers can be selected in the table by clicking on the relevant row.

The button bar provides several function buttons. Using the buttons, you can record data for a serial number or complete a posting:

Designation (name)	Purpose
--------------------	---------

Designation (name)	Purpose
Close	Closes the dialog. Already merged serial numbers remain.
Complete	The currently generated merged batch is completed for the entered/merged serial numbers. The merged batch number is displayed on the terminal. If the user confirms it, the dialog remains open and the user can enter additional data for merged batches.
Merge	Dialog to select and assign serial numbers that are to be merged.
Serial number data	Dialog to enter data for a specific serial number. You can call the dialog for one specified serial number only. The number can also be the number of an already merged serial number from the grid of serial numbers.
Next	Continues with the next workflow step

Dialog "Merge"

In the dialog "Merge", the ingoing serial numbers (of the available merged batches) are identified, grouped and transferred to the superordinate serial number. You must call the dialog for each superordinate serial number and enter the data in the dialog.

You must enter the following data:

Superordinate serial number

A selection list opens where the available, superordinate serial numbers may be chosen. You can identify the superordinate serial number via the [identifier Superordinate SNR = Superordinate](#) in the component list.

Assigned serial number

A selection list opens where you can select the available, ingoing serial numbers. All serial numbers of the registered merged batches are offered that have not yet been assembled and that do not pertain to the superordinate serial number.

1 – n serial numbers can be selected and taken over.

Overview grid

The table shows the selected serial numbers that are to be merged. For each serial number it is obvious if it is a superordinate serial number or an ingoing serial number.

The button bar provides several function buttons:

Designation (name)	Purpose
Cancel	Exits the dialog without modifying, accepting or posting of data.
Accept	If the superordinate serial number or assigned serial number is entered manually in the input field, this function key takes it over in the overview grid.
Delete	The entire assignment is deleted. A new assignment process may be started directly after deletion.
Complete	Directly posts the assignment made.  The assignment is posted on the terminal and server, once it has been completed. Corrections can only be made via the Separating/Rebuilding Serial Numbers dialog.

Dialog "serial number data - batch attributes"

Batch attributes may be entered for the serial number in the "serial number attributes" workflow step of the "serial number data" dialog.

The "attributes" function key allows entering attributes for a batch or for all serial numbers. The input dialog lists all attributes where the option "Capture attribute while generating batch" is enabled for the material type in the [Configuration of the batch attributes](#).

You can use the function key "Continue" to record document links for the serial number.



The list only shows recorded batch attributes. Other available batch attributes are not shown.

Dialog "serial number data – document links"

You can specify document links for a serial number using the dialog "serial number data", workflow step "document links".

The user selects one of the following input options:

- For each serial number

If this option is enabled, the entered document links are only saved for this serial number. The entered document links are stored and suggested the next time a serial number is recorded.

- All serial numbers in this merged batch

If this option is enabled, the entered document links are saved for all serial numbers of this merged batch. The entered document links are stored and suggested the next time when the serial numbers for a merged batch are recorded. It is helpful to enable this option when you start to record document links. This way, all serial numbers in the merged batch include the document links and you need not specify the links for each serial number.

The list only shows the document links recorded in this process. If document links of the serial number are inherited from previous processes, these links are not shown.



You only use this dialog to record document links and not to show the links. You can display all document links that are available for a serial number or a batch/merged batch if you call the batch information (if configured accordingly in the document management).

You can only use this dialog to record document links (URL). Another function is currently not available.

Workflow step - merged batch attributes

You can use the workflow step "merged batch attributes" to record batch attributes for merged batches.

You can use the "attributes" function key to enter attributes for the merged batch that is currently being processed or for this and all future merged batches of this operation. The input dialog lists all attributes where the option "Capture attribute while generating batch" is enabled for the material type in the [Configuration of the batch attributes](#).

You can use the function key "Continue" to record document links for the merged batch.



The list only shows document links recorded, document links that might already be available are not shown.

Workflow step - merged batch documents

You can record document links for a merged batch in the workflow step "document links" of the dialog main view.

The user selects one of the following input options:

- For each merged batch

If this option is enabled, the entered document links are saved for this merged batch only. The entered document links are stored and suggested the next time a merged batch is recorded.

- All merged batches from now on

If this option is enabled, the entered document links are saved for the current merged batch. And the values are saved for all further merged batches (that are generated from now on)

The list only shows the document links recorded in this process. If document links are inherited from previous processes, these links are not shown.



You only use this dialog to record document links and not to show the links. You can display all document links that are available for a serial number or a batch/merged batch if you call the batch information (if configured accordingly in the document management).

You can only use this dialog to record document links (URL). Another function is currently not available.

12 Configuration of Merging Serial Numbers

Activation at machine/workplace

These configurations have to be made for the [machine / workplace](#) to enable the merge of serial numbers:

Parameter name	Value
Workplace configuration → Workplace master data → Workplace category	N Machine
Workplace configuration → Workplace master data → Workplace type	E Single workplace
Workplace configuration → MPL → Batch management	L Batch tracing (input/output batches)
Workplace configuration → MPL → Preceding material buffer	If required, define a preceding material buffer .
Workplace configuration → MPL → Subsequent material buffer	If required, define a subsequent material buffer .
Workplace configuration → MPL → Automat. generation of batch numbers	J Automatic generation of batch numbers for production batches (MPL) enabled

Maintain material types - for the operation

Maintain the [material types](#) to be defined for the operation and adapt them to your specific requirements of data collection.

Maintain material types – for components

Maintain the [material types](#) to be defined in the component list and adapt them to your specific requirements of data collection. Configure at least the following values for the component that is integrated as merged batch including assigned serial numbers in the operation:

Parameter name	Value
----------------	-------

Parameter name	Value
Input batch processing → Inventory management	R - Yes, backflush (retrograde)
General → Options → transferred at interface	Enable this option if you expect consumption postings or final backflushes (notification of goods receipts) for this material type. Then another configuration can be used to control the transfer of merged batches and serial numbers separately.

Maintain reasons

Maintain [reasons for scrap and rework as well as reasons for open quantities](#), if you use these quality classes.

Maintain transport units

Create [transport units](#) in the system if you want to use them for data collection.

Assign material types to transport units

If you want to use [transport units](#) and you created them in the system, you may assign them to the [material types](#) maintained in the system and define a default transport unit for each material type. This one will then be selected in advance in the input dialog.

Perform the assignment in the [assignment of TPU to material type](#).

Activation at the operation

These options have to be set for the operation.

Parameter name	Value
Batch management requirement	Yes
Serial number requirement	U Merge serial numbers

These options are available to edit this information:

Manual maintenance for the operation

Edit the options manually for the [operation](#)

Maintenance of processing code (customizing)

Edit the options for the [processing code](#).

Maintenance of the template (customizing)

Edit the options of the [processing code](#) with value "U" – Merge serial numbers and assign the processing code to an [operation template](#).

Explicit specifications for the interface (recommended procedure)

Explicitly transfer the options for the operation at the interface.

Activation in components list

Maintain the input quantity in the [component list](#) by entering "1.0" units for components listed as merged batch including assigned serials numbers.

Verify whether or not the serial number of one of the incorporated components is to be continued. If this is the case, assign the "superordinate serial number" flag to this component.

Parameter name	Value
Component → Superordinate serial number	F Superordinate

Activation of the link between serial number and input batch

For tracing it is necessary to link the incorporated input batches with the produced serial numbers. To enable this processing, a configuration has to be enabled in [INI configuration](#). The entry itself is part of the default delivery and only needs to be enabled.

Parameter name	Value
INI name	MPL
Section	SERIALNUMBER
Key	CONNECT_SNR_CNR
Value	Y
Active	Enable the entry

Configure batch attributes for data collection when recording serial numbers

If you want to enter batch attributes manually when collecting serial numbers, create the [batch attributes](#) to be recorded in the system in relation to the [material type](#) of the operation. To do so, maintain at least these configurations:

Parameter name	Value
Options → Capture attribute while generating batch	Enable the option.
Options → Position	Specify the position - the system sorts the configured attributes in an ascending numeric order (bottom up).
Data type	Maintain the data type and length of the attribute to be recorded

Control the generation of goods movements

Define for incorporated merged batches and serials numbers as well as for produced merged batches and serial numbers if you require goods movements subject to uploads to be provided. To do so, configure the following settings in [advanced object configuration](#):



The goods movement option has to be enabled for the relevant material type to be able to use this configuration.

Parameter name	Value
Configuration for goods issues (consumptions)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP

Parameter name	Value
Object ID 3	Material type the entry applies for
Object ID 4	CMM_A
Parameter	CREATE_MOVEMENT
Parameter value	Y
Configuration for goods receipts (generated material)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_E
Parameter	CREATE_MOVEMENT
Parameter value	Y

13 Separate/Rebuild Serial Numbers

Usage

This function enables material components listed by serial numbers and already mounted in a component to be exchanged specifically. Mounted material components are demounted and the new material components to be used are mounted. Mounting and demounting are documented in the system and recorded for the relevant serial number.

Procedure of separating/rebuilding serial numbers

In general, there might be two different situations requiring the exchange of components. At the moment, the system does not differentiate between them.

1. Components are demounted from a part while the process is running
An operation to "merge" serial number is running and it turns out that a serial number with a "damaged part" has been mounted. The serial number is recorded for an output batch/merged batch "as usual" and is ready to be demounted/rebuilt (batch status = free, class = yield, quality status = free/open).
2. Components are demounted in a subsequent or separate process step (e.g. rework)
The serial number was recorded for an output batch/merged batch and is ready to be demounted/rebuilt.
Specific identifiers (customer-specific) specify when a serial number gets to this process (e.g. batch status = blocked, class = rework/scrap, quality status = blocked). But by default these indicators are set: (batch status = free, class = yield, quality status = free/open).

In both cases, the function does not require operations or input components to be logged on. The dialog can be carried out at an independent workstation.

Posting of serial numbers

The below-mentioned postings have to be performed for the affected components listed by serial numbers, once they have been rebuilt:

Serial number of the complete, finished component part

This serial number is still assigned to the merged batch from which it was removed.

All relevant data (material number, material type, batch status, class, quality status) remains unchanged.

The connection to the demounted component is deleted. Now it is only included in the history. The connection to the newly mounted part is added accordingly.

Serial number data (attributes, document links) can be entered for the rebuilt part.

Serial number of the demounted component part

This serial number is no longer connected to the entire part from which it was demounted.

For traceability purposes, an entry is made in the batch history (indicating that this component was once mounted in this part).

The serial number of the removed component is not assigned to a new merged batch. It remains as single part (batch with quantity 1) on stock (class = scrap and batch status = blocked).

The merged batch which the demounted serial number derives from also remains unchanged (the inventory is not increased for this batch).

Serial number of the mounted component part

This serial number is taken from a provided merged batch the inventory of which is reduced accordingly.

As in this process no operation is logged on, no merged batch will be logged on as input batch. The inventory of the merged batch is reduced in the background after mounting.

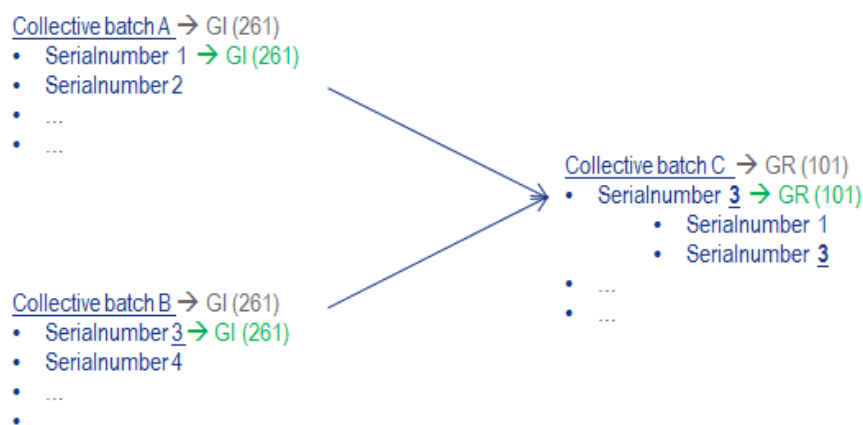
Batch data of the mounted component part is not changed. All relevant data (material number, material type, batch status, class, quality status) remains unchanged.

A connection to the consumed serial numbers/merged batches is established for traceability purposes.

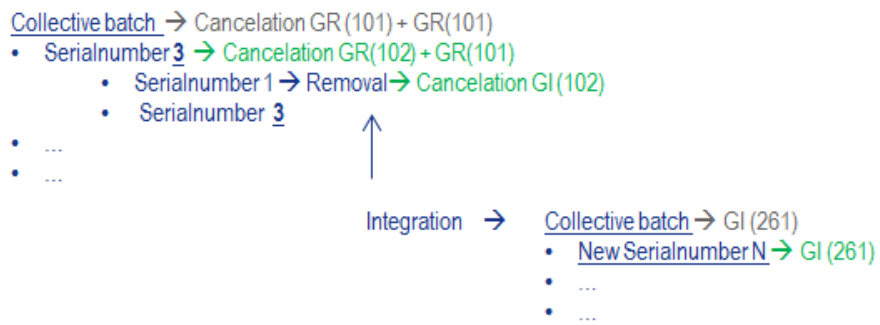
Generation of goods movements

These goods movements are performed for the mounted/demounted components, the complete component part and for the corresponding merged batches.

Goods movements when merging components listed by serial numbers:



Goods movements when separating components listed by serial numbers:



Subject to configuration, the generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches. Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.

14 Separate/Rebuild Serial Numbers at AIP

Usage

The "separate/rebuild serial number" function is used to demount components listed by serial numbers from a component part and to mount a new component as a replacement during assembly processes. The serial number of the component part remains and is continued as a numeric value. All ingoing and outgoing serial numbers are to be identified.

Prerequisite

Various configurations have to be made in the system to use the dialog. Further details can be found in the document dealing with [description of the configuration to separate/rebuild serial numbers](#).

"Separate/rebuild serial numbers" dialog

The "separate serial number" dialog (A_SNR_D) consists of workflows providing extensive possibilities for data collection. By default, the dialog includes the following workflow steps:

Started by the workflow button "serial number data" → Dialog **SNR_DATA_D**

Including the dialogs:

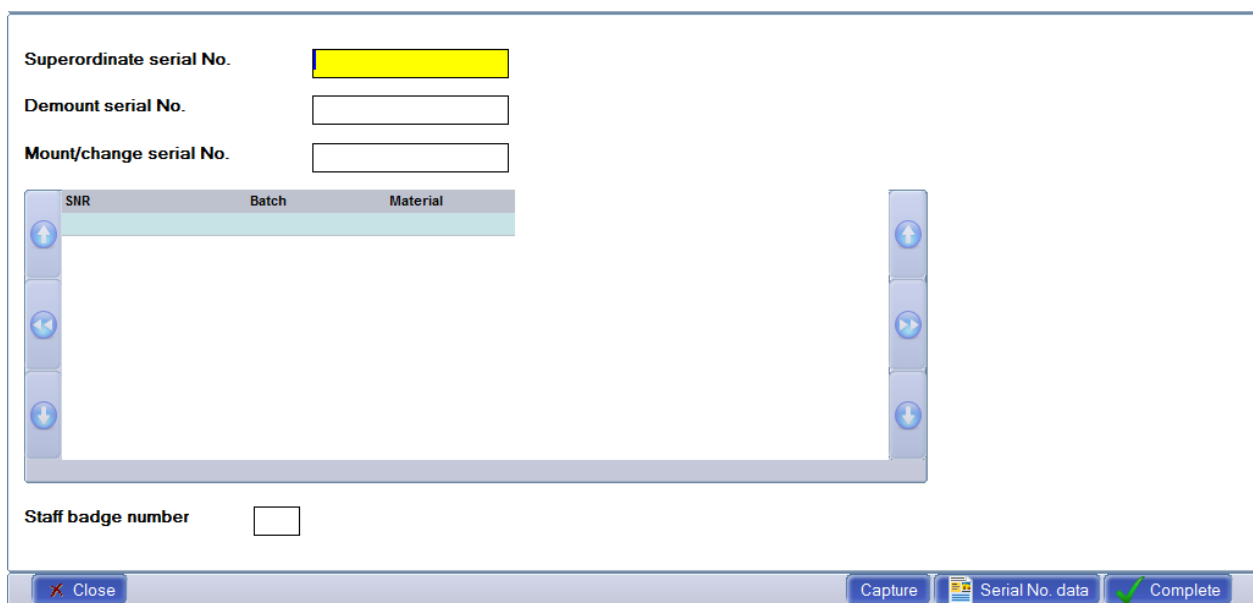
Workflow step/detailed dialog	Technical name	Usage
separate serial number	A_SNR_D	This workflow step shows basic data of the serial number or the user can enter this data here.
Serial number data	SNR_DATA_D	Detailed dialog to enter additional data specific to serial numbers. The detailed dialog can only be started for a single serial number.
Serial number data - attributes	WF_SNR_ATTR_D	Starts the input of attributes for a serial number
Serial number data - document links	WF_SNR_DOC_D	Starts the input of document links for a serial number

Depending on the active workflow step, different function keys are provided.

Basically, data is collected in two steps:

- **Separate/rebuild serial numbers**
At first demounting of a component and mounting of a replacement from/in a component part listed by serial number is documented. The reconstruction is recorded in the system, which leads back to the "separate serial number" workflow step.
The dialog may either be closed or additional data (batch attributes and document links) is entered for the component part listed by serial number.
- **Data collection for the serial number of the finished component part**
Basically, data is collected for the serial number when the serial number is completed.

"Separate/rebuild serial number" dialog



The following data is used in the dialog:

Superordinate serial number

The serial number of the component part from which a component is to be removed has to be entered (mandatory).

Demount serial number

The serial number of the component that is to be removed has to be entered (mandatory).

Mount serial number

The serial number of the component that is to be mounted has to be entered (mandatory).

Staff badge number

Entry of the staff badge number is mandatory.

The result list shows the components currently mounted in the serial number of the component part.

There is a number of function keys facilitating dialog handling, to input data for serial numbers or to complete a posting:

Designation	Usage
Close	Closes the dialog.
Complete	If the user confirms it, the dialog can be continued and remains opened to enter additional merged batches.
Serial number data	Detailed dialog to enter data specific to serial numbers. The detailed dialog can only be started for a single serial number and refers to already mounted serial numbers.

Procedure of the "separate/rebuild serial numbers" function

- The user opens the dialog "separate/rebuild serial numbers" by the function key in the basic terminal screen.
- The user enters the serial number (possibly by scanning) that is affected by separation/rebuilding.
- The result list shows all components mounted in this serial number (through the "merge serial numbers" function).
- The user enters the serial number of the component he/she wants to remove and/or takes it over (by double clicking).
- The user enters the new serial number he/she has mounted.
- Then the user presses the "capture" function key and the serial number is posted along with new component parts.
- The dialog remains opened until the user closes it explicitly. The entered "serial number" and result list remains and/or is still displayed.
- If several component parts need to be rebuilt, the dialog can still be used. Each exchanged component is entered separately.

Detailed dialog Serial number data - batch attributes

Batch attributes may be entered for the serial number in the "serial number attributes" workflow step of the "serial number data" detailed dialog.

The "attributes" function key allows entering attributes for a batch or for all serial numbers. The input dialog suggests all attributes that are assigned the "Capture attribute while generating batch" option in the [configuration of batch attributes](#) regarding the operation's material type.

The "go on" function key allows entering document links for serial numbers.

1 SNR-Attribute 2 Serial No. documents

Serial number

Attribute	Value	Unit	Text
-----------	-------	------	------

Close Attributes Go on

Detailed dialog "serial number data – document links (SNR_DATA_D)

The user may enter document links for each serial number (through [configuration](#)).

The procedure is as follows:

- The user presses the "serial number data" function key in the basic screen of the collection of serial numbers dialog.
- The user enters the document links. The entered/selected document links are displayed as assigned links in the result list.
- The user selects one of the following input options:
 - For each serial number
If this option is enabled, the entered document links are only saved for this serial number. The entered document links are stored and suggested the next time a serial number is recorded.
- Once the user has entered the document links for the serial number, he/she gets to the next workflow of the dialog or back to the basic screen by "go on".

Seriennummer trennen/umbuchen

1 SNR-Attribute
2 Serial No. documents

Serial number

Seriennummer-Dokumentenlink

Link/document

↑

↶

↓

↑

↷

↓

X Close

Add
Delete
↶ Back
✓ Complete

CTAIP V2.0.3.4 Server.scc7.4 TNR:131
04.07.14 07:53:58

15 Configuration of Separating/Rebuilding Serial Numbers

Dialog configuration

Define the AIP function key in the file *ctaipbut.ini*:

```
[ANR-LN-Page2]
...
1=A_SNR_D,L,SNR trennen, SNR_trennen.png
```

The below-mentioned sections are required in the layout configuration of *ctaisplay.ini*:

```
[A_SNR_D_GRID.LST ]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=

DLL=C20,110,L,SNR
ALIAS LEER2=(DUMMY1)=C1,10,L
CNR=C20,0,L,CNR
ALIAS LEER3=(DUMMY2)=C1,40,L
SAPCNR=C20,110,L,Charge
ALIAS LEER4=(DUMMY3)=C1,10,L
ATK=C20,110,L,Material

[WF@DOC_DATA_D]
FILTER=
SECTION=DOC_DATA_D.LST
DATAFIELDS=
FILE=doc_data_d.lst
AUTOFILTERCOL=
MODE=DATALOCKUNTILSHOW=TRUE|

[DOC_DATA_D.LST]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=

LINK=C20,400,L,link/document

[WF@ATTR_DATA_D]
SECTION=ATTR_DATA_D.LST
DATAFIELDS=
FILE=attr_data_d.lst
AUTOFILTERCOL=
MODE=DATALOCKUNTILSHOW=TRUE|
```

```
[ATTR_DATA_D.LST]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=
```

```
ATTR=C20,100,L,Attribute
VALUE=C20,80,L,Value
EINH=C20,60,L,Unit
TEXT=C20,200,L,Text
```

System configuration

Collection of batch attributes

If you want to enter batch attributes manually when collecting serial numbers, create the [batch attributes](#) to be recorded in the system in relation to the [material type](#) of the ready-mounted serial number. To do so, maintain at least these configurations:

Parameter name	Value
Options → Capture attribute while generating batch	Enable the option.
Options → Position	Specify the position - the system sorts the configured attributes in an ascending numeric order (bottom up).
Data type	Maintain the data type and length of the attribute to be recorded

Control the generation of goods movements

Define for incorporated merged batches, serials numbers and produced serial numbers if you require goods movements subject to uploads to be provided. To do so, configure the following in [advanced object configuration](#):



The goods movement option has to be enabled for the relevant material type to be able to use this configuration.

Parameter name	Value
----------------	-------

Configuration for goods issues (consumptions)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_A
Parameter	CREATE_MOVEMENT
Parameter value	Y
Configuration for goods receipts (generated material)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_E
Parameter	CREATE_MOVEMENT
Parameter value	Y

16 Variants of Batch Grouping

Summary

Batches are grouped and/or combined due to different situations:

- Not relevant to the process
It is assumed here that the identified material only switches positions and this movement is irrelevant to batch tracing (e.g. picking or transfer posting of batches to a group and provision for production).
- Relevant to the process
This function facilitates posting processes allowing to log on many input batches at once to a machine/OP, guide them through the process and to repost them afterwards to produced output batches for traceability purposes.

Variants

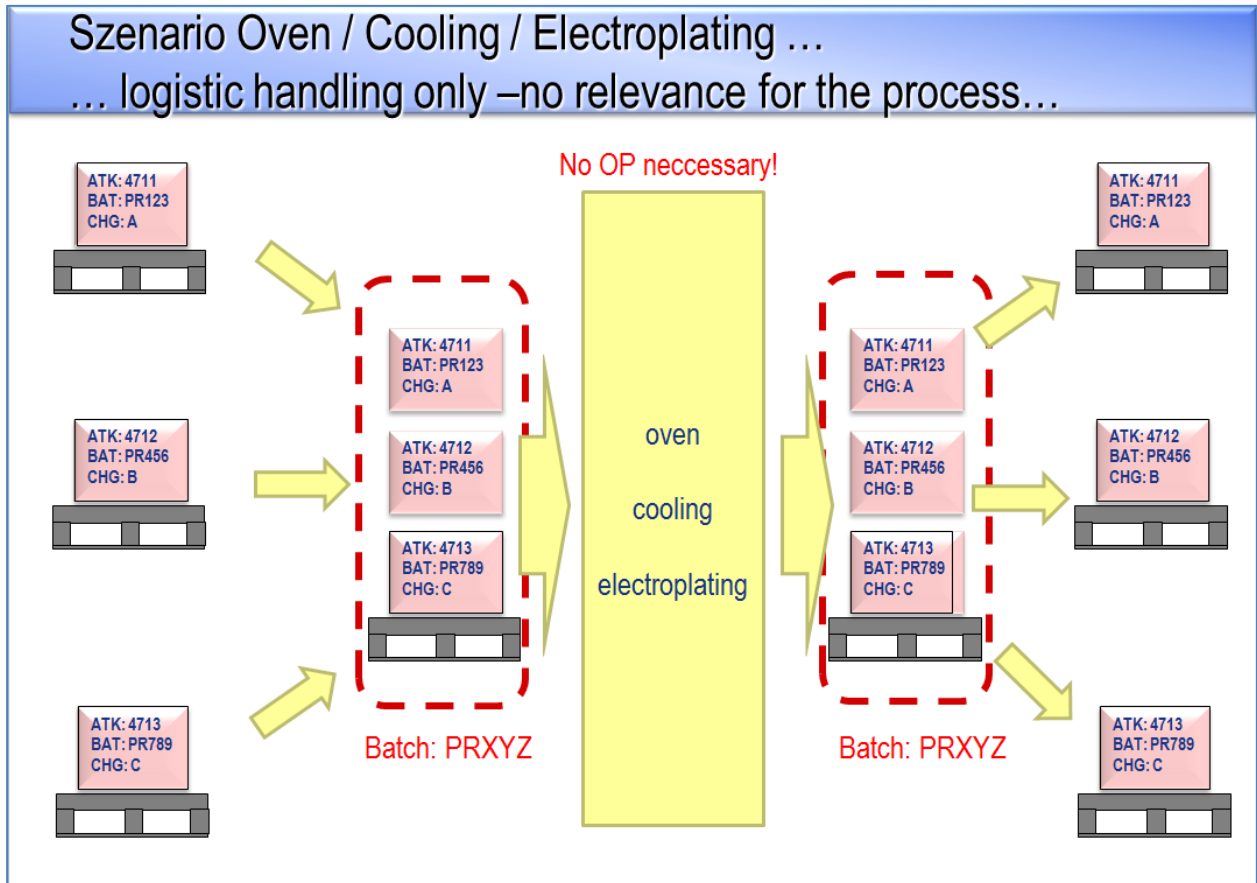
In general, there are the following two variants to group batches.

- Illustration: irrelevant to process --> generate/cancel batch group
- Illustration: relevant to process --> batch grouping

Generate/cancel group batch (not relevant to process)

General

The function is used every time when it is necessary to combine batches of different or identical materials, which is to be resolved/canceled at a later point in time providing the batches "unaffected".



Prerequisites, restrictions

- Original batches remain and are not "consumed" by grouping.
- During grouping, original batches cannot be processed or, for example, archived somewhere else.
- No OP is required - groupings may be performed at any point in time.
- Batches may still be removed unless the grouping has been created finally.
- A label can be printed for the generated grouping.
- Groupings may only be cancelled altogether. Individual batches cannot be removed.
- All group members are affected if the grouping is reposted (change of material buffer).
- Tracing does not consider the grouping.

AIP procedure

There are two steps:

- Generate group batch
- Cancel group batch

Results

The following results can be expected after generating the batch group:

- Batch characteristics
 - Individual batches still keep their material number, material type, material buffer, etc.
- Batch status
 - Once the "generate group batch" function has been executed, the single batches are in the "running" status. The group batch is assigned the "running" status.
- Goods movements
 - No goods movements are posted for the single batches or the group batch.
- Traceability
 - The group of batches is not relevant to tracing.
- History
 - The group of batches is documented in the history of individual batches.

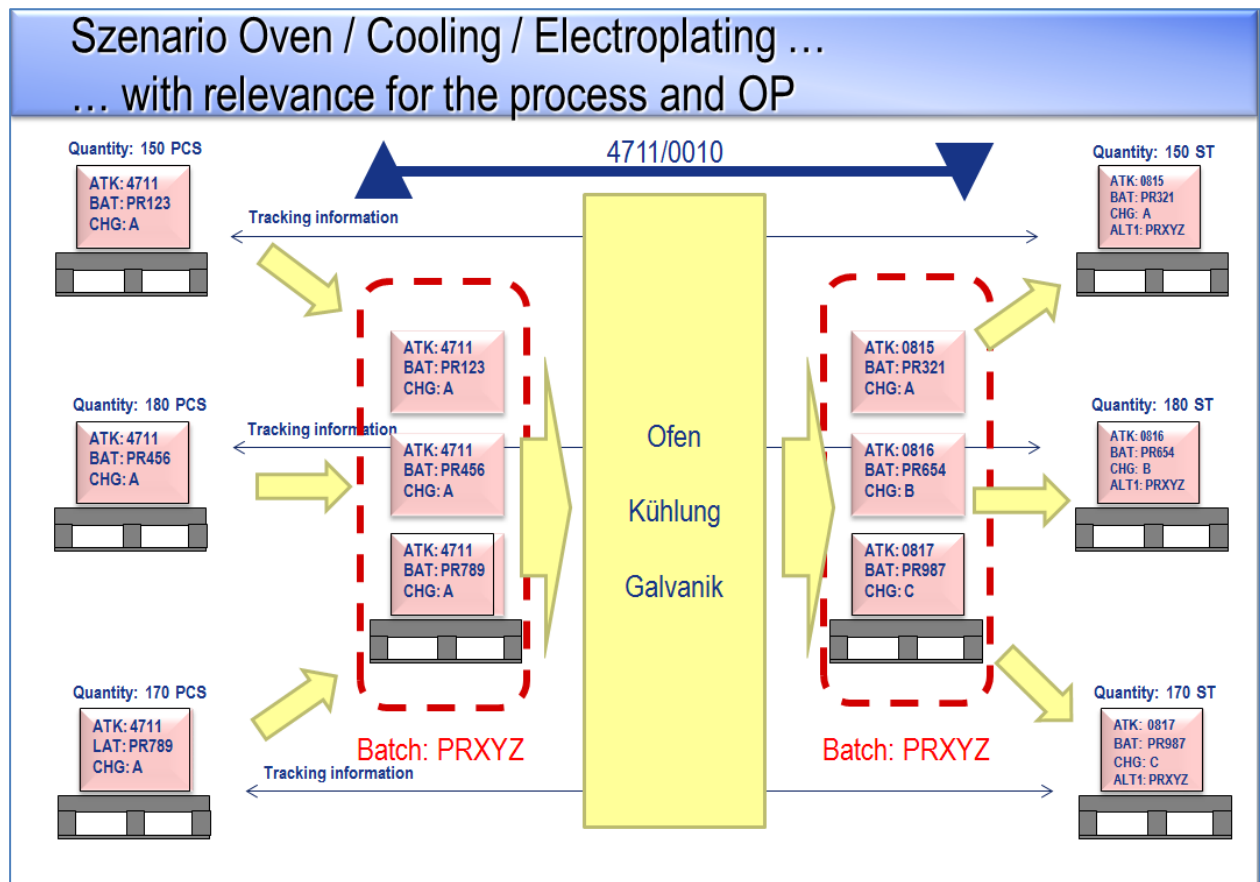
The following results can be expected after canceling the batch group:

- Batch characteristics
 - Individual batches still keep their material number, material type, material buffer, etc.
 - The group batch has taken over the following values from a "system batch" (copy template):
 - Material number
 - Material type
 - Material buffer
 - The quantity of the group batch is the total amount of all assigned individual batches and, therefore, is assigned the unit "PCE/pieces".
- Batch status
 - Once the "cancel group batch" function has been executed, the single batches are in the "free" status. The group batch is assigned the "processed" status. The "grouping" no longer exists.
- Goods movements
 - No goods movements are posted for the single batches or the group batch.
- Traceability
 - Cancelling group batches is not relevant to traceability.
- History
 - Cancelling the group batches is documented in the history of individual batches.

Batch grouping (relevant to process)

General

The function is used every time when batches of different or identical materials are used in a process and output batches are finally changed for all ingoing batches at a point in time.



Prerequisites, restrictions

- This function can only be used if grouping is directly performed "within the process". Groupings cannot be performed beforehand as part of a preceding picking process.
- The machine has not been configured as MPL machine.
- Input batches are "consumed" by grouping.
- During grouping, input batches cannot be processed or, for example, archived somewhere else.
- Operations are required: an OP appropriate for processing is searched via the order network or by reservations affecting all orders.

AIP procedure

The AIP procedure is described here.

Results/ processing

The "batch grouping" dialog (U_GROUPING) provides several functions.

- Add batch, start OP: CNRGRP.ADD
- Remove batch from list: CNRGRP.REMOVE:
- Complete group: CNRGRP.UNLOAD:

After executing the single functions, processing is performed in the dialog U_GROUPING.

Add batch/ start OP (CNRGRP.ADD)

This function adds the entered batch. To do so, the OP for the batch is logged on to the machine.

Inspections:

- General batch inspection:
 - The batch is available
 - The batch has the status "free"
- Automatic search for the operation matching the entered batch
 - OP is searched via the order network and batch assignment
 - The batch was generated in an OP, which is provided as piece of information for the batch. Due to the order network, the system knows the next OP to be processed within the order. The appropriate OP can be found by the order network.
 - If no OP is found via the network, it is searched using batch reservations.
 - An error message appears if no OP is found.
 - If an OP is found, it will only be started, unless it is already running at the machine.
- Further inspections:
 - Checking if the OP is planned for the machine and/or machine group.
 - Checking if the batch is reserved for another OP.
 - Checking if the batch can be logged on to the component.
 - An error message occurs if no component is found.
 - The component must be identical to the material number of the batch, otherwise an error message occurs.
- An error message appears if the admissible number of OPs running at the machine is exceeded.
- Start OP:

- The OP is started and set to the "running" status.
- The batch is reserved for the OP.

Remove batch (CNRGRP.REMOVE):

This function is used to cancel the reservation of the OP at the batch.

Inspections:

- The batch is available. Otherwise, an error message occurs.

Unload batch group (CNRGRP.UNLOAD):

This function is used to log on the reserved batches.

Inspections:

- Are there any running OPs including reserved batches at the machine?

Posting:

- Determination of all reserved batches
- Log output batch and input batch on.
 - At first generate a new batch number for the output batch
 - The new output batch is logged on with the reserved input batch.
- Log off output batch and input batch
 - The input batch is logged off
 - The output batch is logged off
 - The configured, alternative batch numbers and attributes are saved for the output batch.
 - The connection between input batch and output batch is saved for tracing purposes.
- This process is repeated for all batches included in the group.

17 Create/Cancel Group Batch

Summary

Within the production area, it is necessary to compress batches to a group with one ID and to provide this package/group within another area. It is assumed here that the identified material only switches positions and this movement is irrelevant to batch tracing (e.g. picking or transfer posting of batches in a group and provision for production or transport).

Usage

The function is used to combine batches of different or identical materials, which is to be resolved/canceled at a later point in time providing the batches "unaffected".

Prerequisite/configuration

The configuration for AIP is described [here](#).

The configuration for AIP2 is described [here](#).

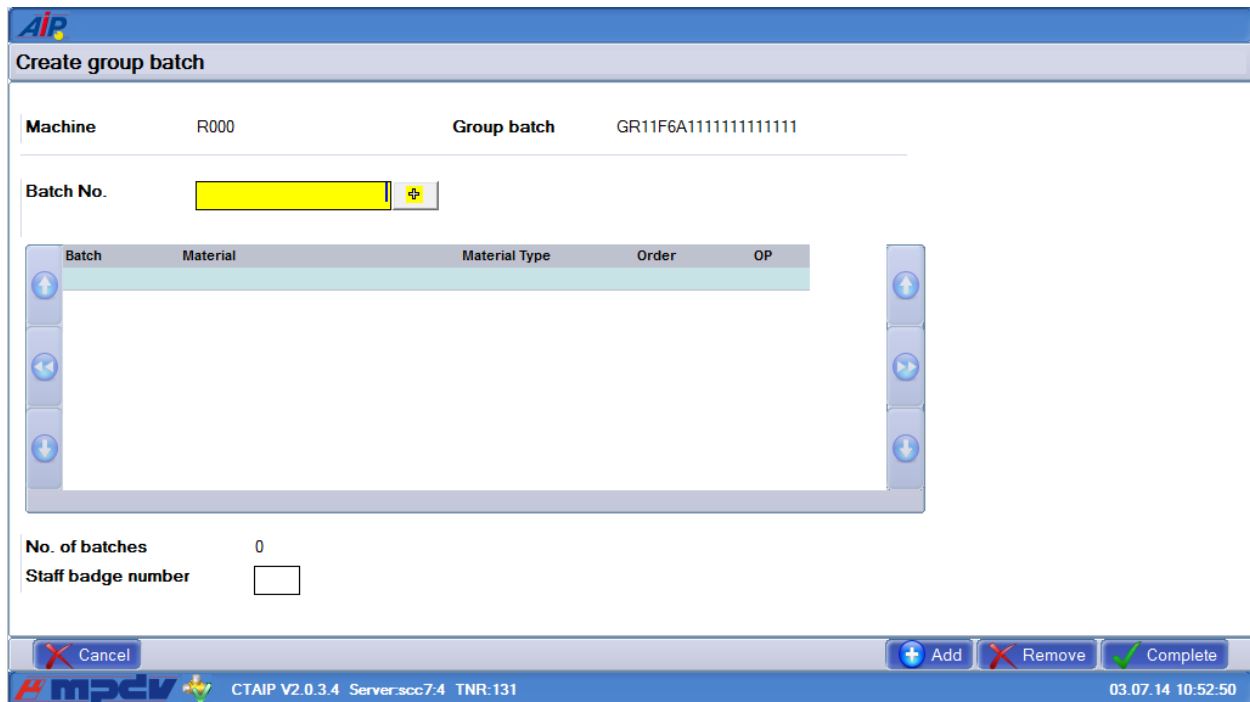
The logical process and posting are described [here](#).

Generate group batch



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog



Description of display fields:

- **Machine**
Used workplace/machine
- **Group batch**
Generated group batch that is currently being created.
- **Batch number**
Entered number of the batch that is to be recorded for the group batch.
- **Number of batches**
Number of individual batches included in the group batch.
- **Staff badge number**
The user's staff badge number.
- **Result list**
List of all individual batches included in the group batch with batch number, material number etc.

Function keys

- **Function key "cancel"**
The function key terminates the dialog and rejects data input.
- **Function key "add"**
An individual batch may be assigned to the group batch using this function key.
- **Function key "remove"**
A selected individual batch may be removed from the group batch using this function key.
- **Function key "complete"**

This function key completes the group batch, i.e. it is generated.

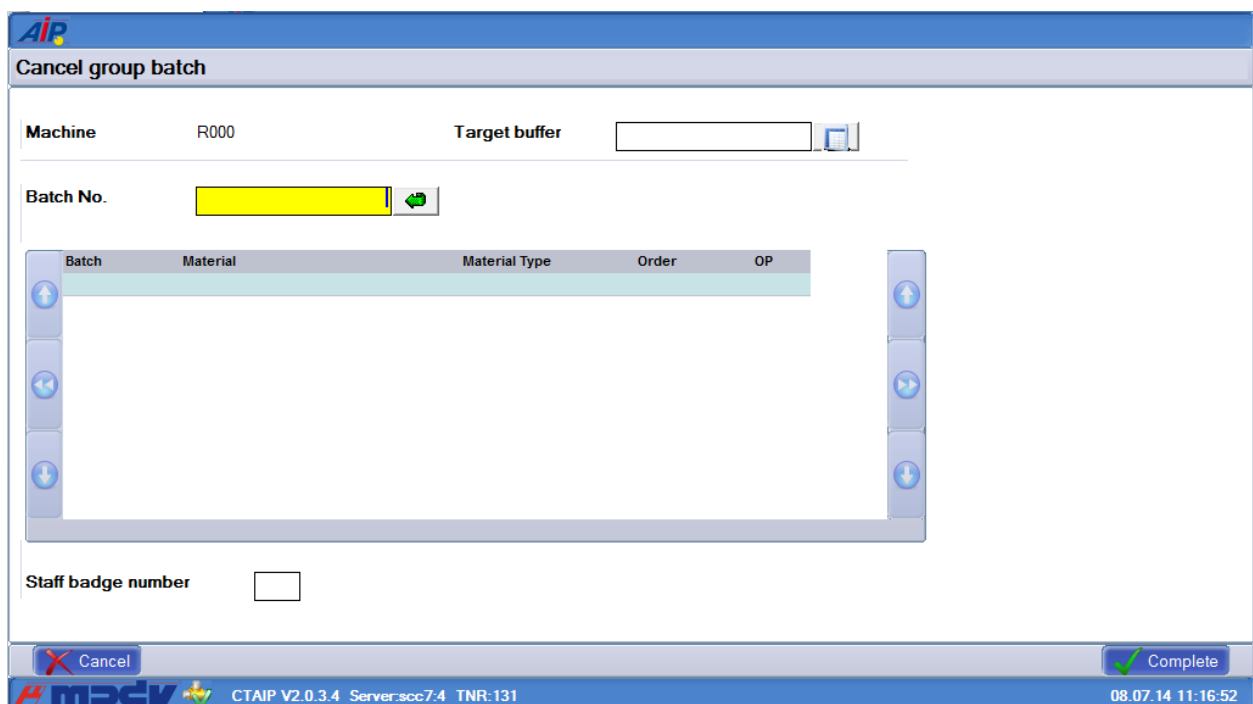
Procedure

For the user the procedure is as follows:

- The user opens the "create group batch" dialog in the basic screen of the terminal
- The user enters the batches (manually or by scanning) he/she wants to group
- The user may view the group batch number in the dialog
- The dialog's result list shows the entered batches that are to be grouped.
- Individual batches may still be removed or added, as long as the user has not confirmed the dialog by the "complete" key.

Cancel group batch

Dialog



Description of display fields:

- **Machine**
Used workplace/machine
- **Target buffer**
Material buffer to which the batch is transferred. If the field remains empty, no transfer posting takes place and the material buffer remains.
- **Batch number**

Entered number of the group batch that is to be canceled (re-posted).

- **Staff badge number**

The user's staff badge number.

- **Result list**

List of all individual batches included in the group batch with batch number, material number etc.

Function keys

- **Function key "cancel"**

The function key terminates the dialog and rejects data input.

- **Function key "complete"**

This function key cancels the group batch.

Procedure

For the user the procedure is as follows:

- The user opens the "cancel group batch" dialog in the basic screen of the terminal
- The user enters the group batch (manually or by scanning) he/she wants to cancel
- The dialog's result list shows the batches pertaining to the grouping.
- In the "material buffer" field the user may enter a new material buffer to which all individual batches of the group are to be re-posted.
 - If the user does not fill out the field, the material buffers of the individual batches remain as before.
- As long as the user has not closed the dialog by "cancel group", it remains open and/or can be cancelled. Data will be rejected.

18 Create/Cancel Group Batch

Usage

The function can be used to combine several batches to a group batch at the AIP and to cancel it at a later point in time.

Dialog configuration

Dialog configuration (CNR.BATCH GROUP A)

Define the AIP function keys in the file *ctaipbut.ini*:

```
[ANR-IN-Page4]

...

1=BATCH_GROUP_A,L,Gruppenlos bilden,Losgruppe_bilden.png
2=BATCH_GROUP_C,L,Gruppenlos auflösen,Losgruppe_auflösen.png
```

The sections [BATCH_GROUP_A.LST] and [BATCH_GROUP_C.LST] are required in the layout configuration of *ctaisplay.ini*:

```
[BATCH_GROUP_A.LST]

GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

CNR=C20,100,L,Batch
ATK=C20,100,L,Material
ATKBEZ=C40,140,Designation
HZTYP=C40,150,L,Material Type
RESAUNR=C12,100,L,Order
RESAGNR=C4,50,L,OP
; for set then position
;U_POSITION=C8,60,R,Position

[BATCH_GROUP_C.LST]
```



```
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

CNR=C20,100,L,Batch
ATK=C20,100,L,Material
ATKBEZ=C40,140,Designation
HZTYP=C40,150,L,Material Type
RESAUNR=C12,100,L,Order
RESAGNR=C4,50,L,OP
; for set then position
;U_POSITION=C8,60,R,Position
```

System configuration

Further system configurations are not required to group/cancel batches.

19 Batch Grouping

Summary

In production it might be necessary to process batches at once in one process (e.g. heating furnace, conditioning system, washing machine). To this end, the user logs on the affected batches within a group to AIP.

Usage

To the user the OP is not important to record times and after processing, users do not want to post every single output batch in the system. Therefore, this function basically facilitates the posting process for the user.

Prerequisite/configuration

The configuration is described [here](#).

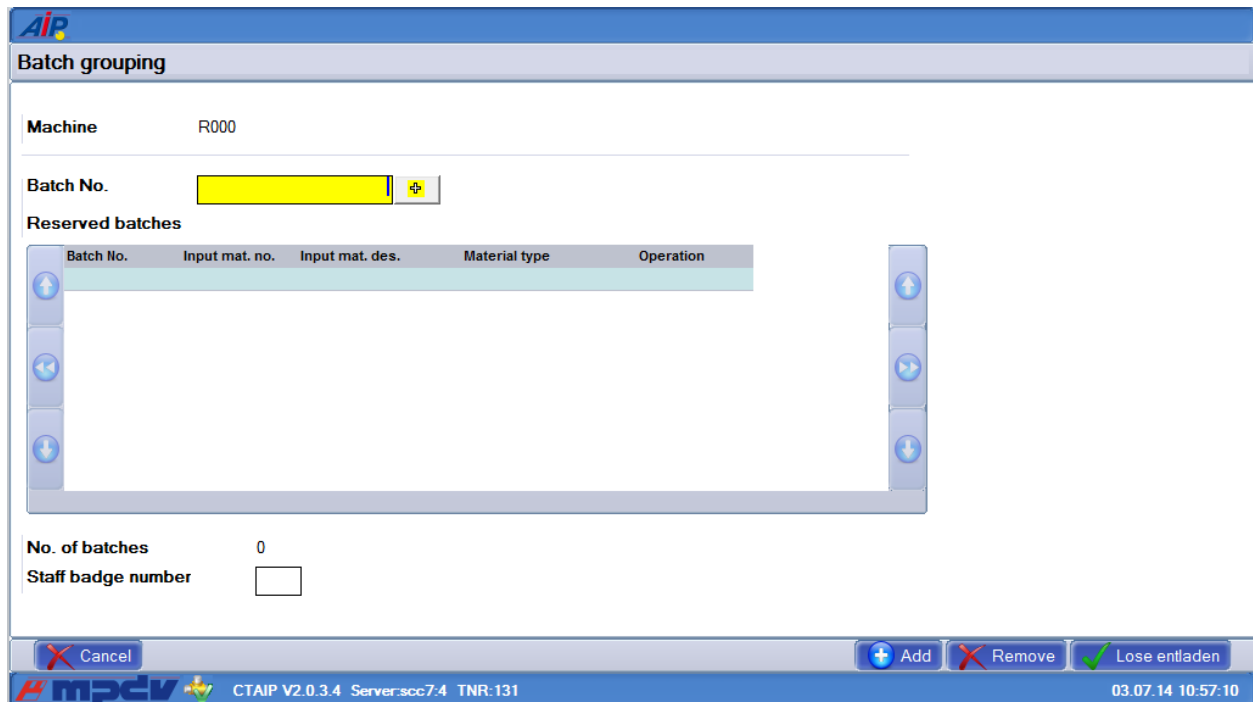
The logical process and posting are described [here](#).

Batch grouping



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog



Description of display fields:

- **Machine**
Used workplace/machine
- **Batch number**
Entered number of the batch that is to be recorded for the group.
- **Staff badge number**
The user's staff badge number.
- **Result list "reserved batches"**
List of all individual batches included in the group with batch number, material number etc.

Function keys

- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "add"**
An individual batch may be added to the group using this function key.
- **Function key "remove"**
A selected individual batch may be removed from the group using this function key.
- **Function key "unload batch"**
This function key cancels the group and output batches are changed automatically for all input batches and operations registered in the background.

Procedure

For the user the procedure is as follows:

- The user opens the "batch grouping" dialog in the basic screen of the terminal
- The user enters the batch (manually/by scanning) he/she wants to add
- The user presses "add" and confirms the entered batch.
- The entered batch is assigned to the group. The batch is reserved for the OP.
- The user enters further batches he/she adds to the group.
- The user closes the dialog unless he/she wants to add further batches.
- To cancel the group and/or when completing the process, the user opens the dialog and presses the "unload batch" key. Consequently, the relevant registered operation is logged off and finished for every included input batch and an output batch is completed.

20 Batch Grouping

Usage

You make use of the function to use several batches of a group within the process at AIP.

Dialog configuration

Dialog configuration (BATCH_GROUP_B)

Define the AIP function keys in the file **ctaipbut.ini**:

```
[ANR-LN-Page4]
...
3=BATCH_GROUP_B,L,Losgruppenbildung,Losgruppierung.png
```

The section [BATCH_GROUP_B.LST] is required in the layout configuration of **ctaipay.ini**:

```
[BATCH_GROUP_B.LST]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

CNR=C20,100,L,
ATK=C20,100,L,
ATKBEZ=C40,140,
HZTYP=C40,150,L,
RESANR=C12,100,L,Operation
```

System configuration

Further system configurations are not required to group batches.

An appropriate OP has to be available in the system to be able to use the batches in the dialog and in order for an OP to be logged on in the background.



The relevant OP can be determined by:

- a special batch reservation (affecting several orders)

- the order network (sequence of OPs in an order)

21 Decision on Changing Input Batch

Usage

For traceability reasons, every time an input batch is changed, a new merged batch/output batch is to be produced for material listed by serial numbers.

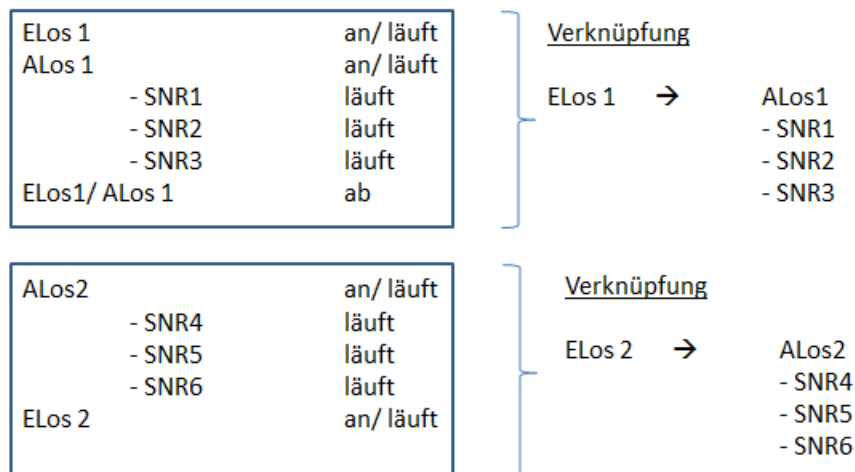
The function "decision on changing the input batch" is used every time when input batches are changed and it has to be ensured that the reported serial numbers deriving from different input batches are not mixed up in one output batch.

Example:

Two input batches from different lots are used for one material. Input batches are changed, once the first lot of the input batch has been consumed. The material/input batch of the first lot is logged off and the material of the second lot is logged on as input batch. There must be at least two different merged batches/output batches, as the serial numbers of two different input lots or input batches are assembled.

The "decision on changing the input batch" function and the relevant posting procedure for input batch changes make sure the right connections are generated between input and output batches and the user is forced to "pay attention" and to post all serial numbers before logging an input batch off.

Example:



Prerequisite/configuration

The warning message can be enabled and/or disabled. The configuration is described [here](#).

Results/inspections

The user is forced to make a decision when changing input batches:

- Log input batch off --> No
- Log input batch off --> Yes

The input batch is logged off by the system if the user decides to log off the input batch.

The dialog is executed and the input batch to be logged off is posted accordingly:

- Change the batch status (processed, free, blocked)
- Material movement (consumption --> goods issue (261))
- Connections between the output batch (merged batches and subordinate batches) and the input batch are completed in batch tracing. The system has already established a connection between the input batch to be logged off and the currently running output batch when the output batch was generated.

If this connection is not required, an option of the material type "output batch connection = J/N" makes sure the connection is deleted.

- The batch history is updated.

The user makes an entry (staff badge number) for the logged off input batch and the logged in input batch in the batch history. Consequently, one can trace back who decided to log on a new input batch and to ignore the serial numbers mounted with the old input batch. These serial numbers are only posted with the new input batch, although they have not been produced by it.



This function can be applied in the "merge serial numbers" dialog and to "change output batches". The text has to be adjusted specific to the project.

22 Decision on Changing Input Batch

Summary

For traceability reasons, every time an input batch is changed, a new merged batch/output batch is to be produced for material kept with serial numbers. The function "Decision on changing the input batch" is provided for this purpose.

Usage

This function is used when input batches are changed and it has to be ensured that the reported serial numbers deriving from different input batches are not mixed up in one output batch.

Decision on changing the input batch - AIP warning



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog

Step 1: Change input batch

Change input batch <CE_WL_MPL>

Workplace R000 Material 453-9971

Operation SNRAUNR90010

Input batch

POS Input mat. no.	Input mat. des.	Custom. batch no.	Remains Unit
1 453-9970	453-9970	SAM100	5 ST

Staff badge number

Cancel Post batch

CTAIP V2.0.3.4 Server:sc7:4 TNR:131 03.07.14 10:50:18

Step 2: Decision on changing the input batch

The user receives a message at the terminal asking whether the batch to be changed is actually to be logged off or if there are still remaining serial numbers to be entered. The dialog text can be customized.

Procedure

For the user the procedure is as follows:

- The user opens the "input batch change" dialog by pressing "input batch change" in the basic screen of the terminal.
- Step 1: Change input batch
 - The user selects the input batch to be logged off and/or enters the new input batch to be logged on and uses the "post batch" key.
- Step 2: Decision on changing the input batch

The user receives a message at the terminal asking whether the batch to be changed is actually to be logged off or if there are still remaining serial numbers to be entered.
- Step 3: User decision
 - The user wants to enter further serial numbers for the running input batch and clicks "No". The user has to enter his/her staff badge number.
 - The user decides to confirm the message by clicking "yes". Consequently, the input batch is logged off.

23 Configuration of Decision on Changing Input Batch

Usage

The function "decision on changing the input batch" is used when input batches are changed and it has to be ensured that the reported serial numbers deriving from different input batches are not mixed up in one output batch.

For this reason, the "decision on changing the input batch" alert may be configured as an intermediate dialog when changing input batches.

System configuration

Material type:

These options have to be set for the material type:

- Input batch processing > decision on changing the input batch = ON
If this option is enabled, the alert is shown in the "decision on changing the input batch" dialog.
- Output batch processing > delete batch assignment = ON
Using this option deletes the most recent assignment between input batch and current output batch. This deletion is necessary as the assignment is established immediately after the input batch has been logged on.

24 Splitting Batches

Summary

The "split batch" function can be used to divide an existing batch into several individual batches. The following "batch types" may be split:

- "simple" batch (split off quantities with new batch numbers)
- merged batch (split off subordinate batches/serial numbers into new merged batches)

Prerequisite

In general, all batches to be split off have to be available.

The "split batch" function can only be applied to batches meeting the below-mentioned conditions:

- only batches in the batch status "free" may be split
- only batches assigned the batch class "yield" may be split

Variants

In general, there are the following two variants to split batches.

- [Splitting batches at AIP](#)
AIP provides a function to split the entered batches/merged batches into several batches/merged batches. A label may be printed each for the batches/merged batches split off.
- [Splitting batches at MOC](#)
The MOC batch data overview provides a function to split the entered batches/merged batches into several new batches/merged batches.

Result

The below-mentioned results can be expected after splitting batches at AIP or MOC:

Splitting "simple" batches:

- the batches split off include the entered quantity
- the batches split off are assigned the status "free"
- If no remaining quantity is available, the old batch is in the "free" or "processed" status.
- The old batch has the batch class "yield"

- the batch number of the old batch:
 - is a new batch number
 - is the original batch number

Splitting merged batches:

- the merged batches split off include the selected subordinate batches/serial numbers and the relevant quantity.
- the merged batches split off are assigned the status "free"
- If no remaining quantity is available, the old merged batch is in the "free" or "processed" status.
- the old merged batch is assigned the "yield" batch class and includes all subordinate batches/serial numbers that have not been split off and/or no subordinate batches/serial numbers if all of them were split off.
- the batch number of the old merged batch:
 - is a new batch number
 - is the original batch number

25 AIP Batch Splitting

Summary

Some production processes (e.g. goods issue) require a batch (e.g. with large quantities) to be split into several batches (e.g. with smaller quantities) to be able to provide materials in smaller bins/containers at machines/workplaces. With this function the terminal provides an opportunity to separate batches and to print a label for the new batches.

Usage

These split variants are possible:

- "Simple" batches
When splitting "simple" batches, the user separates the required quantities from the original batch and generates new batches for these split quantities.
- Merged batches
When splitting merged batches, the dialog shows the individual sub-batches/serial numbers assigned to the merged batch. The user selects the sub-batches/serial numbers he/she wants to split off to generate a new merged batch.

Prerequisite

The relevant configuration has to be made to be able to use this function.

Terminal functions

The function is triggered manually by using the key of the "split batch" dialog (BATCH_SPLIT).



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog/start

The dialog opens by input of the batch number/merged batch number. The batch information on the batch/merged batch is shown.

Batch information	
Batch	1902FASS09
Material	4711
Designation	22402
Type	MAT
Class	G
Operation	C140000510000006
Machine	4202_00
Material buffer	MELT_OUT
Remaining qty.	10,0 KG
Prod.date	10.12.2013 14:54:26
Status	F
Note	
Merged batch	N

Cancel
 Splitten

Splitting "simple" batches

AIP

Split batch

Batch No. 1001234
Class G
Remaining qty. 61,0 ST

Quantity ST
Reason

☒ Yield
☐ Scrap
☐ Rework
☐ Open quantity

Quantity	Class	Reason
10,000	G	0
10,000	A	1
5,000	H	0
4,000	O	0
10,000	G	0

☐ Repost remaining quantity to new batch
☒ Reduce batch to remaining quantity

Cancel
 Add
 Remove
 Split

CTAIP V2.0.3.4 Server:scg7:4 TNR:131
03.07.14 11:40:58

Description of display fields:

- **Batch number**
Entered number of the batch that is to be split.
- **Class**
Batch class of the batch to be split.
- **Remaining quantity**
Remaining quantity of the batch to be split. At the beginning, it is the original quantity.

- **Quantity**
Quantity to be split off. The user enters this quantity.
- **Class**
Batch class of the quantity to be split off. This can be:
 - Yield
 - Scrap
 - Rework
 - Open quantity
- **Reason**
Reasons for the selected batch class.
- **Repost remaining quantity to new batch**
The remaining quantity of the original batch is reposted to a new batch.
- **Reduce batch to the remaining quantity**
The original batch is reduced to the remaining quantity.
- **Result list**
The result list shows all batches split off including the respective quantity, batch class and reason.

Function keys

- **Function key "add"**
The result list shows the entered quantity as separate entry (later the batch split off).
- **Function key "remove"**
The concerned row of the result list is selected and can be removed. The envisaged quantity is not split off and assigned again to the original batch.
- **Function key "split"**
When using the function key, all entries of the result list are split off including the relevant quantity from the entered batch.
- **Function key "cancel"**
This function key deletes all entries and the dialog is closed.

Procedure:

- The user opens the "split batch" dialog in the basic screen of the terminal
- The user selects the batch he/she wants to split off.
- Batch data of the entered batch is displayed.
- The user enters the split quantity, batch class and the reason, if necessary.
- The "add" function key adds the quantity and/or batch to be split off to the result list and displays it.
- The user may split off additional batch quantities

- The result list shows all quantities/batches to be split off
- The user decides if the remaining quantity of the original batch is to be reposted to a new batch or if it is to remain with the original batch.
- The user closes the dialog by the "split" function key and the split quantities will be posted.

Splitting merged batches

Description of display fields:

- **Batch number**
Entered number of the merged batch that is to be split.
- **Class**
Batch class of the merged batch to be split.
- **Remaining quantity**
Remaining quantity of the merged batch to be split. At the beginning, it is the original quantity.
- **Class**
Batch class of the merged batches to be split off. This can be:
 - Yield
 - Scrap
 - Rework
 - Open quantity
- **Reason**
Reasons for the selected batch class.
- **Display list**
The display list shows all sub-batches/serial numbers including the relevant quantity and article number assigned to the entered merged batch. Individual sub-batches/serial numbers are selected and split off into a new merged batch.

Function keys

- **Function key "reload"**
The display list is updated.
- **Function key "reverse"**
The selected sub-batches/serial numbers are unselected.
- **Function key "split"**
The selected sub-batches/serial numbers are assigned to a new merged batch and removed from the existing merged batch.
- **Function key "cancel"**

The dialog is closed by this function key.

Procedure:

- The user opens the "split batch" dialog in the basic screen of the terminal.
- The user selects the batch he/she wants to split off.
- Batch data of the entered batch is displayed:
 - Material
 - Quantity
 - Class
 - Reason
 - Batch status
- If it is a "merged batch" (indicator at the batch), the result list shows the serial numbers/subordinate batches assigned to this merged batch.
- The user chooses (multiple selection) the serial numbers/sub-batches he/she wants to split off into a new merged batch.
- The user selects the class for the new merged batch (and thus for the newly assigned serial numbers/sub-batches)
- The function key "split" generates the new merged batch with the split off serial numbers. The list of serial numbers is reduced and additional merged batches may be generated and, as a result, serial numbers can be split off.
- The remaining serial numbers/sub-batches remain with the original merged batch (existing number).
- Then the user closes the dialog.

Result

Result when splitting "simple" batches:

- A new batch is generated for the quantity split off.
- The batch quality depends on the selection made
- Subject to the selected quality (yield/scrap), the batch status of the new batch is --> released/blocked
- The quantity of the old batch is reduced by the quantities split off. If the quantity is zero, the batch status of the old batch is "processed".
- Other data of split batches, such as the material number, designation, material type, storage location, material buffer, PPS batch is taken over from the original batch.
- The split event is displayed in the batch history (old batch and new batches).
- The graphic/tabular overview shows which batches have been split off from a batch.

Result when splitting a merged batch:

- The new merged batch is generated and includes all sub-batches/serial numbers selected for splitting
- The quality is identical for the merged batch and sub-batches/serial numbers (as defined)
- The quantity included in the new merged batch is the total quantity of all sub-batches/serial numbers
- Subject to the selected quality (yield/scrap), the batch status of the new merged batch and the sub-batches/serial numbers is --> released/blocked
- The old merged batch is no longer connected to the sub-batches/serial numbers split off
- The quantity of the original merged batch has been reduced by the quantity of the sub-batches/serial numbers split off. If the quantity is zero and/or if there are no longer sub-batches/serial numbers, the batch status of the old batch is "processed".
- Other data of sub-batches/serial numbers and the new merged batch, such as the material number, designation, material type, storage location, material buffer, PPS batch are taken over from the original batch.
- The split event is displayed in the batch history (old merged batch and new merged batches + serial number).
- The graphic/tabular overview shows which merged batches have been split off from a merged batch.



This dialog does not allow entering attributes/document links for individual serial numbers/sub-batches pertaining to the merged batch.

26 Splitting Batches at AIP

Usage

You use the function to split up a batch into several batches at AIP. Subject to the batch type, batches may be split in the following ways:

- Splitting off quantities into individual batches
- Splitting off individual batches from existing merged batches into a new merged batch

Dialog configuration

Dialog configuration (CNR.SPLIT)

Define the AIP function keys in the file ***ctaipbut.ini***:

```
[ANR-LN-Page3]
...
3=BATCH_INFO,L,Los splitten,Los_splitten.png
```

The sections [BATCH_S_SPLIT] and [BATCH_SPLIT] are required in the layout configuration of ***ctaisplay.ini***:

```
[BATCH_S_SPLIT]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

GRID_CELLPAINT=ON
EXAMINE_CELLBKCOLOR=SELECT,SELECT,X-clLime

SELECT=C1,50,Z,*
DLL=C20,150,Z,batch
ATK=C20,150,L,article
SGR:REST=N12.0,100,R,quantity
EINH=C3,60,Z,unit
```

```
;CKL=C1,150,Z,class
```

```
[BATCH_SPLIT]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
GRID_CELLPAINT=ON
```

```
EXAMINE_CELLBKCOLOR=CKL,CKL,G-clLime|A-clRed|N-clBlue|O-clYellow
```

```
EGR=N12.3,250,R,quantity
```

```
CKL=C1,150,Z,class
```

```
EGG=C4,100,Z,reason
```

System configuration

Further system configurations are not required to split batches.

27 Merging Batches

Summary

The "merge batches" function can be used to merge existing batches into one common batch. The following "batch types" may be combined:

- "simple" batch and "simple" batch (adding up quantities)
- merged batch and merged batch (grouping of assigned subordinate batches/ serial numbers)

Prerequisite

In general, all batches to be merged have to be available.

The batch merge function can only be applied to batches meeting the below-mentioned conditions:

- Only batches with the same material (material number) can be merged
- Only batches with the same material type can be merged
- All batches must have a positive remaining quantity (batch class = yield, remaining quantity > 0). Consequently, scrap batches cannot be merged.

The following combinations are neither supported nor allowed:

- combination of "simple" batches and merged batches
- combination of an individual batch/serial number and a merged batch
- Combination of an individual batch/serial number and an individual batch/serial number

Variants

In general, there are the following two variants to merge batches.

- [Merge batches at AIP](#)
AIP provides a function to merge the entered batches/merged batches into a new batch/merged batch. A label can be printed for the merged batch.
- [Merge batches at MOC](#)
The MOC batch data overview provides a function to merge the entered batches/merged batches into a new batch/merged batch.

Result

The below-mentioned results can be expected after merging batches at AIP or MOC:

Merging "simple" batches:

- The merged batches have a remaining quantity of 0
- The merged batches are assigned the status "processed"
- The new batch has the status "free"
- The new batch has the batch class "yield"
- The batch number of the new batch:
 - is a new batch number
 - is a batch number pertaining to the merged batches

Grouping merged batches:

- the grouped merged batches have a remaining quantity of 0 and do no longer include subordinate batches/serial numbers
- the combined merged batches are assigned the status "processed"
- the new merged batch has the status "free"
- the new merged batch is assigned the "yield" batch class and includes all subordinate batches/serial numbers of the combined merged batches
- The batch number of the new merged batch:
 - is a new batch number
 - is a batch number pertaining to the combined merged batches

28 AIP Batch Merge

Summary

In production partly consumed containers with the same material are often combined to one container. However, the system considers the materials to be located in different batches. With this function the terminal provides an opportunity to merge batches and to print a label for the new batch.

Usage

At the moment, it is impossible to:

- combine "normal" batches with merged batches and/or serial numbers
- combine merged batches and serial numbers or to add several serial numbers to one merged batch. The merged batch number including the serial number always has to be entered.
- combine individual serial numbers into one merged batch. Their merged batch number always has to be entered. It is also possible to use the standard function "collection of serial numbers" A_SNR with the relevant configuration.

Prerequisite

The relevant configuration has to be made to be able to use this function.

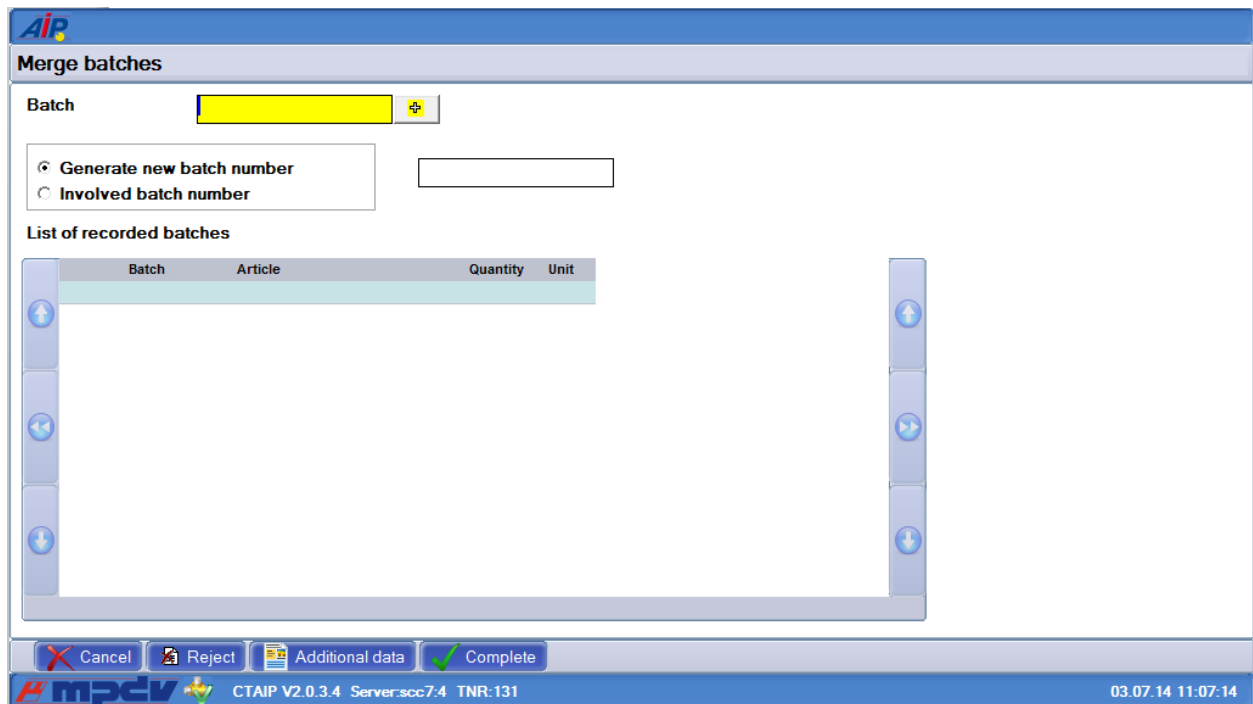
Terminal functions

The function is triggered manually by using the key of the "merge batches" dialog (BATCH_MERGE).



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog



Description of display fields:

- **Batch**
Batch number to be entered by the user (manually/by scanning)
- **Generate new batch number / involved batch number**
Specifies whether the new batch is to be generated with a new batch number or if an involved batch number is used for the new batch. If an involved batch number is used, this one has to be entered.
- **List of recorded batches**
Shows the entered batches that are to be merged.

Procedure for the "merge batches" function:

- The user opens the "merge batches" dialog by the relevant function key in the basic screen of the terminal
- In the "batch" field the user enters the affected batches (manually or by scanning) he/she wants to merge. Data may be entered manually or by scanning. If batches are entered manually, the input will be confirmed after each batch by . This empties the field for the next input.
- The user specifies whether the new batch is to be generated with a new batch number or if an involved batch number is used for the new batch. If an involved batch number is used, this one has to be entered.
- The "list of recorded batches" shows the batches the user has already entered and wants to merge.

- The user applies the "complete" key. Then the "batch" field and the "list of recorded batches" are emptied. The user receives an intermediate message with the generated batch number for the merged batch. Additional batches may be merged.
- The dialog remains opened until the user closes it explicitly by pressing "cancel". Then batches are merged into a new batch.
- A label can be printed for the merged (new) batch.

Function keys

- **Function key "cancel"**
This function key rejects the entered data and/or empties the result list. New data may be entered. The dialog remains open.
- **Function key "complete"**
Entered batches are merged into one batch. By using this function key, the dialog is emptied (i.e. no data included) and the new batch (batch number) is shown on the terminal. A label is printed subject to configuration.
- **Function key "reject"**
Entered batches are not merged into one batch. The data input is rejected.
- **Function key "additional data"**
Subject to configuration, this function leads to the next workflow to record attributes and/or document links.

Result

- **By the "cancel" function**
The dialog remains open. All entered data are rejected. The input data are not posted. Data input can be restarted.
- **By the "complete" function**
This function key closes the dialog. Entered batches are merged into one batch and data are posted accordingly. The amount of merged batches is reduced accordingly and set to the "processed" batch status. The new batch includes the merged amount. If merged batches are combined, the serial numbers will be assigned to the new merged batch. Label printing may be triggered by this.
- **By the "enter" function**
Entered data are recorded for each batch but not yet posted.
This function key records the individual batches once they have been entered and displays them in the result list. The dialog remains open.
- **By the "additional data" function**

Similar to the output batch change, it is possible enter attributes and document links for each combined batch/merged batch. If they are configured, relevant workflows are integrated for data collection. Attributes and document links from previous batches are not taken over.

This dialog does not allow entering attributes/document links for individual serial numbers/sub-batches pertaining to the merged batch.



In addition, the event is also shown in the batch history (old batches and new batch and old merged batches and new merged batch and serial number).

The graphic/tabular overview shows which batches have been merged into one batch.

29 Merge Batches at AIP

Usage

You use the function to merge several batches to one batch at AIP. Subject to the batch type, batches may be merged in the following ways:

- Merging simple batches and/or their quantities to one batch
- Merging individual batches pertaining to an already existing merged batch to one new merged batch

Dialog configuration

Dialog configuration (CNR.SUMMARIZE)

Define the AIP function keys in the file *ctaipbut.ini*:

```
[ANR-LN-Page3]
```

```
...
```

```
2=BATCH_MERGE,L,Lose zusammenfassen,Lose_zusammenfassen.png
```

The sections [BATCH_MERGE] and [DOC_BATCH_MERGE.LST] and [ATTR_BATCH_MERGE.LST] are required in the layout configuration of *ctaisplay.ini*:

```
[BATCH_MERGE]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
EXAMINE_SCANEXPR1=SLOS=J
```

```
EXAMINE_SCANCOLOR1=clBlue
```

```
DLL=C20,150,Z,batch
```

```
ATK=C20,150,L,article
```

```
SGR:REST=N12.0,100,R,quantity
```

```
EINH=C3,60,Z,unit
```

```
[DOC_BATCH_MERGE.LST]

GRID_FONT=Arial

GRID_FONTSIZE=9

GRID_COLOR=clBlack

GRID_BACKGROUND=clWhite

LINK=C20,400,L,link/document
```

```
[ATTR_BATCH_MERGE.LST]

GRID_FONT=Arial

GRID_FONTSIZE=9

GRID_COLOR=clBlack

GRID_BACKGROUND=clWhite

;GRID_ORDER=DAT=- | ZEI=- | SNR
```

```
ATTR=C20,100,L,attribute

VALUE=C20,80,L,value

EINH=C20,60,L,unit

TEXT=C20,200,L,text
```

System configuration

Further system configurations are not required to group batches.

30 Advanced Batch Information

Summary

Various batch information needs to be made available to manufacturing processes.

Usage

The function is used to display the information stored to the batch. This refers to:

- General batch information
- Batch quantities
- Batch attributes
- Alternative batch numbers



The dialog is structured in workflows. These workflows can be started by clicking on the individual tabs or by clicking "next" in each individual workflow.

Prerequisite

Batch attributes have to be configured for the batch to be able to display them.

Alternative batch numbers can only be displayed if they are used in specific projects.

Show general batch information

Dialog



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Batch information

1 Batch information
2 Quantities
3 Attributes
4 ...

Batch No. 1001234

Material XX
Designation Diffusor Rohr
Type ROHR
Class G

Operation
Machine 50500
Material buffer MP_V_1168

Prod.date 04.03.2013 00:00:00
Status F
Q-Status O
M-Status U

Note beschichtet

Cancel
 Go on

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03.07.14 12:52:40

Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Material**
Material number of the batch
- **Designation**
Material name of the batch
- **Type**
Material type of the batch
- **Class**
Batch class (yield, scrap, rework, open quantity)
- **Operation**
Operation that has produced the batch or which it is logged on to.
- **Machine**
Machine/workplace that has produced the batch or which it is logged on to.
- **Material buffer**
Material buffer that includes the batch.
- **Manufacturing date**
Manufacturing date of the batch
- **Status**
Batch status
- **Q-status**

Quality status of the batch.

- **M-status**

Material status of the batch.

Function keys

- **Function key "cancel"**

This function key terminates the dialog and rejects data input.

- **Function key "go on"**

This function key leads the user to the next configured workflow of the dialog (quantities, attributes, alternative batch numbers).

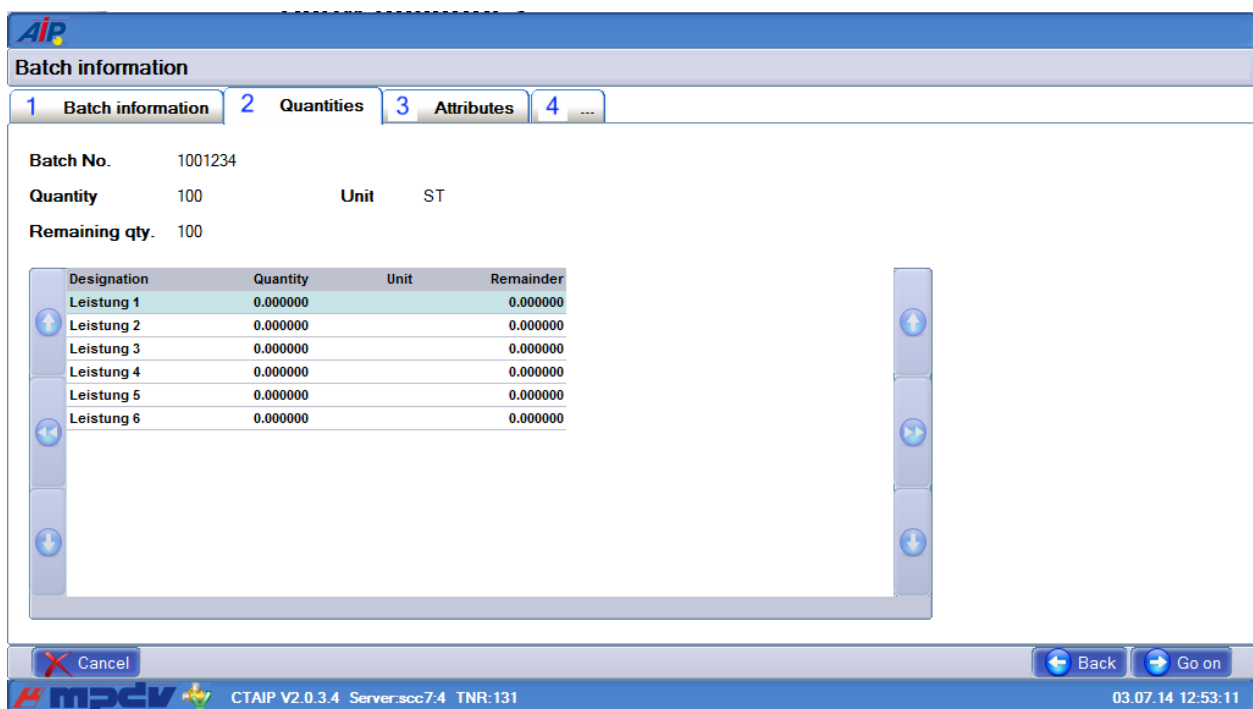
Procedure

For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.
- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.
- The user applies the "batch information" workflow.

Show quantities

Dialog



Designation	Quantity	Unit	Remainder
Leistung 1	0.000000		0.000000
Leistung 2	0.000000		0.000000
Leistung 3	0.000000		0.000000
Leistung 4	0.000000		0.000000
Leistung 5	0.000000		0.000000
Leistung 6	0.000000		0.000000

Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Quantity**
Initial quantity of the batch.
- **Remaining quantity**
Current remaining quantity of the batch.
- **Unit**
Batch unit
- **Activity (1-6)**
Activity fields of the batch (to convert into other units)

Function keys

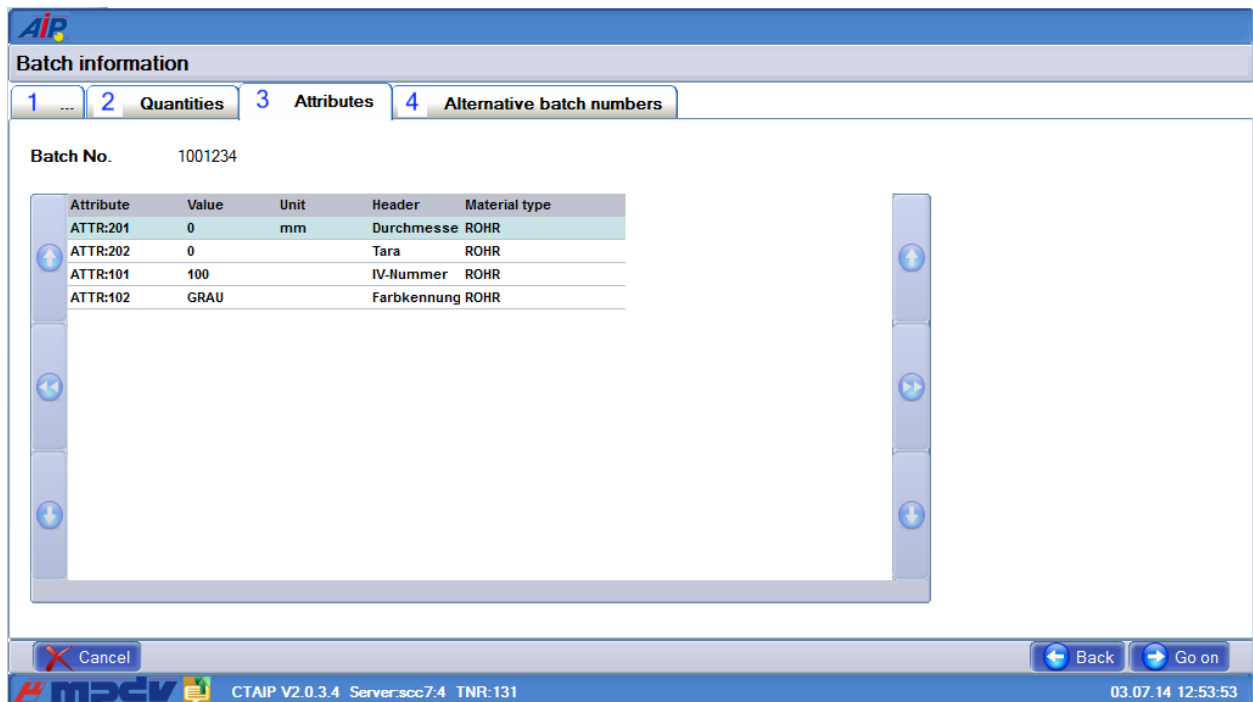
- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "go on"**
This function key leads the user to the next configured workflow of the dialog (attributes, alternative batch numbers).
- **Function key "back"**
This function key leads the user back to the previously configured workflow of the dialog (batch information, quantities, attributes, alternative batch numbers).

Procedure

For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.
- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.
- The user applies the "quantities" workflow.

Show batch attributes**Dialog**



Attribute	Value	Unit	Header	Material type
ATTR:201	0	mm	Durchmesse	ROHR
ATTR:202	0		Tara	ROHR
ATTR:101	100		IV-Nummer	ROHR
ATTR:102	GRAU		Farbkennung	ROHR

Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Result list: attributes**
Shows the configured batch attributes for the material type of the batch including value and relevant captions. The displayed columns are configurable.

Function keys

- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "go on"**
This function key leads the user to the next configured workflow of the dialog (attributes, alternative batch numbers).
- **Function key "back"**
This function key leads the user back to the previously configured workflow of the dialog (batch information, quantities, attributes, alternative batch numbers).

Procedure

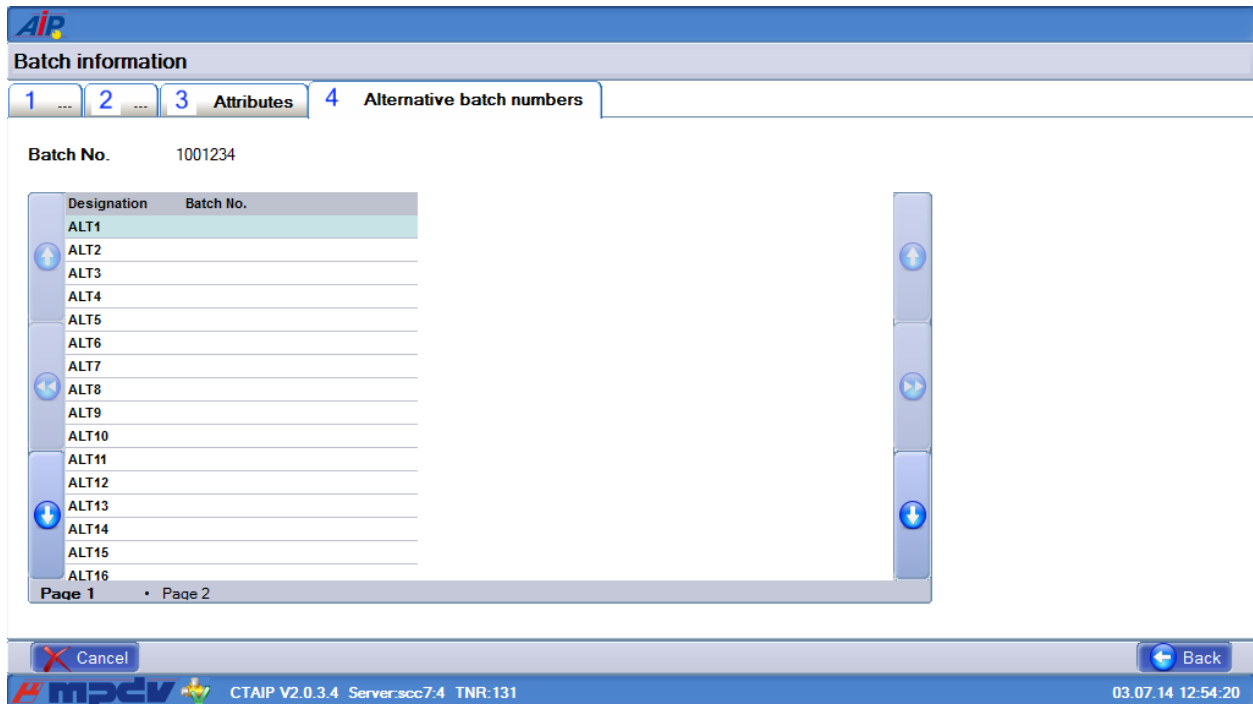
For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.
- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.

- The user applies the "attributes" workflow.

Show alternative batch numbers

Dialog



Batch information

1 ... 2 ... 3 Attributes 4 Alternative batch numbers

Batch No. 1001234

Designation	Batch No.
ALT1	
ALT2	
ALT3	
ALT4	
ALT5	
ALT6	
ALT7	
ALT8	
ALT9	
ALT10	
ALT11	
ALT12	
ALT13	
ALT14	
ALT15	
ALT16	

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Cancel Back

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Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Result list: alternative batch numbers**
Shows the alternative batch numbers for the material type of the batch including relevant value. Alternative batch numbers are optional and entered for specific projects.

Function keys

- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "back"**
This function key leads the user back to the previously configured workflow of the dialog (batch information, quantities, attributes, alternative batch numbers).

Procedure

For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.

- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.
- The user applies the "alternative batch numbers" workflow.

31 Advanced Batch Information

Usage

You use the function to view information on a batch at the AIP terminal. Advanced batch information includes:

- General batch information
- Quantities
- Batch attributes
- Alternative batch numbers

Dialog configuration

Dialog configuration (LOS_INFOS)

Define the AIP function key in the file **ctaipbut.ini**:

```
[ANR-LN-Page3]
```

```
1=LOS_INFOS,L,Losinformation,Shipping Box Closed Information.png
```

The sections [LOS_INFOS_ATTR.LST] and [LOS_INFOS_QUA.LST] and [LOS_INFOS_ALTER.LST] are required in the layout configuration of **ctaipay.ini**:

```
[LOS_INFOS_ATTR.LST]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
KENNUNG=C20,100,L,Attribute
```

```
ATTR_VAL=C20,80,L,Value
```

```
EINH=C20,80,L,Unit
```

```
TXT=C20,80,L,
```

```
MATTYP=C40,140,
```

```
[LOS_INFOS_QUA.LST]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
KENN=C20,100,L,
```

```
ALIAS LEER1=(DUMMY1)=C1,30,L
```

```
MENGE=C20,80,R,
```

```
EINH=C4,40,R,
```

```
ALIAS LEER2=(DUMMY1)=C1,30,L
```

```
REST=C20,80,R,
```

```
[LOS_INFOS_ALTER.LST]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
ALT=C20,100,L,
```

```
CNR=C20,200,L,
```

System configuration

Further system configurations are not required to display advanced batch information.

32 Weighing Components

Usage

For operations that are not subject to batch management, it is possible to record batches in relation to discrete material components. In this special case, batches may be entered in relation to the charge via a special terminal function.

This entry function replaces the collection of quantities for the operation (e.g. partial upload) and records material consumption in relation to material components. This consumption is also posted as material movement in the system.

Prerequisites

- The function can weigh more than one charge. In this case, the operation's secondary quantity has to include the number of charges. The additional data fields of the components and operation regarding the weighing function (e.g. target quantity per charge, tolerances, etc.) also need to be taken into account.
- Every time an operation is changed, the target quantity for each charge is recalculated (formula: $\text{target quantity per charge} = \text{primary target quantity} / \text{secondary target quantity}$). It is neither possible to set the value for the target quantity per charge manually nor to display it on MOC. If the secondary target quantity is not set, the value 1 is assumed by default and only one charge is processed in the weighing operation.
- In this case, the machine does not allow to enter automatic quantities additionally.
- All components have to be managed by the "weight" unit (kilogram).



The entry function cannot be used offline.

Configuration

The configuration document describes the configurations required to use and enable the entry function at the terminal.

Posting

Data for label printing

The schema **a_vbrkomp** is provided in initial data for label printing.

This data can be printed on the label during weighing (dialog action KEW_RECORD):

Acronym	Type	Length	Description
DLG.MNR	C	10	Machine
DLG.ANR	C	40	Operation
DLG.ATK	C	40	Material number
DLG.SLP	C	40	BOM item
DLG.CHARGE	C	20	Batch
DLG.CST	C	1	Batch status F/S (free/blocked)
DLG.SGR:GUT	DEC		Target quantity
DLG.SGE:GUT	C	3	Unit of the target quantity
DLG.EGR:GUT	DEC		Actual quantity
DLG.EGE:GUT	C	3	Unit of the actual quantity
DLG.EGR:MENGE	DEC		Input quantity
DLG.EGE:MENGE	C	3	Unit of the input quantity
DLG.DAT	DATE		Date of the posting
DLG.ZEI	ZEIT		Time of the posting
DLG.KNR	C	10	The reporting person's badge number

This data can be printed on the label when completing the charge (dialog action KEW_ABSCHLUSS):

Acronym	Type	Length	Description
DLG.MNR	C	10	Machine
DLG.ANR	C	40	Operation
DLG.ATK	C	40	Material number
DLG.ATKBEZ	C	40	Material designation
DLG.CHARGE	C	20	Batch
DLG.EGR:GUT	DEC		Quantity of the batch
DLG.EGE:GUT	C	3	Unit
DLG.DAT	DATE		Date of the posting
DLG.ZEI	ZEIT		Time of the posting
DLG.KNR	C	10	The reporting person's badge number

Scale interfacing

The weight can also be entered in the weighing dialog by means of a connected scale. Upon opening the KOMP_WIEG dialog, the value is requested from the scale via the PCC driver interface and entered in the "input quantity" field.

This is an example for entering the scale value in the INI file of the driver:

<WAAGENTREIBER>.INI
V:WAAGE:NETTO=Nettogewicht_Waage

If an OPC interfacing is used, changed scale values can be sent automatically by the OPC server. For this purpose, the below-mentioned parameter has to be entered in the file "OPCMPDV.INI"
SETVALEVENTS=V:WAAGE:NETTO

<WAAGENTREIBER>.INI
SETVALEVENTS=V:WAAGE:NETTO ← additional entry
V:WAAGE:NETTO=Nettogewicht_Waage

33 Weighing of Components

Summary

Some production areas require components to be weighed using scales and to provide them for further production processes.

Usage

The function is always used when the user prepares charges using the relevant material quantities. To save time, users often weigh several charges at once.

The logical process and posting are described [here](#).

Prerequisite/configuration

The configuration is described [here](#).

Weighing components

Components for a charge are weighed by the AIP dialog A_VBRKOMP.



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

The "weighing" function at the terminal provides a list of required input materials (discrete material components). The component-related data collection can be opened from this list.

Once the operation to be weighed has been logged on, the "weighing components" dialog can be started by the "weigh" function key from the toolbar of the basic screen of the terminal. After opening the dialog, the terminal automatically generates a new batch for the order. This is presented as batch in the following dialog.

Please note: The dialog only opens if "discrete" material components (consumption type = D) exist for the running operation.

The actual weighing process of the relevant components is performed by the "weigh charge" function in a detailed dialog (see below).

Dialog

AIP
<A_VBRKOMP>

Workplace W2030
Operation M30002000010
Batch PR4RH6B11K

Material 4529-42
Designation

Status
☒ Free
☐ blocked

Components

POS	Article	Target qty.	Actual qty.	Remaining qty.	UOM	Designation	Tolerance	Dev
1	9901	169,000	0,000	0,000	L	Ethylalkohol	0,000	
2	3011	3380,000	0,000	0,000	G	Phosphorsäure	0,000	
3	4011	152,100	0,000	0,000	G	Glyzerin	0,000	
4	8511	16731,000	0,000	0,000	L	Destilliertes Wasser (Aquadest)	0,000	

Staff Number

Cancel
Weigh comp.
Quantity adjustm.
Reject charge
Complete charge

04.07.14 10:37:22

Workplace

Weighing workplace to which the operation to be weighed is logged on.

Operation

Weighing order/operation

Batch

Batch number for the charge that is currently to be produced.

Material

Material number for the charge that is currently to be produced.

Status

Batch status for the charge that is currently to be produced after weighing. This can be:

- Free
- Locked

Staff badge number

The user's staff badge number.

Components list

The component list shows all components of the currently registered weighing operation that are relevant for the charge to be produced. These details for the component are displayed to perform weighing:

- Article number (material number of the component from the component list)
- Target quantity
- Actual quantity (quantity entered upon weighing, at first 0,000)
- Remaining quantity (computed remaining quantity after weighing, at first 0,000)
- Designation (component name from the component list)
- Tolerance (calculated, admissible tolerance of the component/component list)
- Deviation (calculated, admissible deviation of the component/component list)

Procedure:

The user selects the first component that is to be weighed and starts weighing.

- The relevant component row is highlighted in **green** if the component is within the variance tolerance (at the component, percentage).
- Materials falling short of the variance tolerance are highlighted in **red**.
- Materials the weighing result of which exceeds the variance tolerance but is still within the tolerances of quantity adjustment (of the component) are highlighted in **blue**.
- The row is shown in black font, provided that the component has not yet been weighed.
- If the weight value is beyond the variance tolerance but within the tolerance of quantity adjustment, posting will be accepted but the user has to adjust quantities for the other components (automatically in the system). The input dialog cannot be closed if this modification is not made.

Quantities have to be entered for all input materials included in the component list.

Functions

The paragraphs that follow describe the functions provided in the input dialog to weigh components:

"Weigh charge"

The "weigh charge" function opens the dialog to weigh a selected individual component (KOMP_WIEG).

Weigh charge				<KOMP_WIEG>	
Workplace	W2030		Batch	PR4RH6P113	
Operation	M30002000010		Component	4529-42	
			BOM item	1	
Component batch	PR4RD1D113				
Default quantity	169,000	L			
Actual qty.	150,000	KG			
Balance	0,000	KG	Allowed deviation	0,000	%
Input quantity	0,000	KG	Actual deviation	100,000	%
Staff Number	9999				
Cancel			Weigh charge		

mpciv 18.07.14 09:09:15

The current status of target and actual quantity is presented after selecting a component that is to be weighed. The user has to enter a relevant batch for consumption posting in the "component batch" input field. The batch has to be available in HYDRA with the appropriate material and the status "free" and it must have a remaining quantity ≥ 0 . The dialog can be closed by entering the weighing quantity and the staff badge number (optional).

Successful weighing has the following additional effects on the system:

- Indicators are recorded for the component
- The weighed quantity is added to the actual quantity of the component
- Material movements are generated for consumption (261)
- A batch assignment is generated for the batch $\Leftrightarrow$ component batch.



- The article of the batch always has to match the article of the component.
- A component may also be weighed several times.
- The amount (stock) of component batches is only reduced if the relevant material type is assigned the "retrograde inventory collection" flag.

Once the weight has been entered, the displayed actual quantity and remaining quantity are updated.

The displayed component requirements are updated in the table once the component has been weighed successfully.

<A_VBRKOMP>

Workplace W2030

Operation M30002000010

Batch PR4RH6B11K

Material 4529-42

Designation

Status

☒ Free
 ☐ blocked

Components

POS	Article	Target qty.	Actual qty.	Remaining qty.	UOM	Designation	Tolerance	Dev
1	9901	169,000	150,000	19,000	L	Ethylalkohol	0,000	
2	3011	3380,000	150,000	3230,000	G	Phosphorsäure	0,000	
3	4011	152,100	0,000	0,000	G	Glyzerin	0,000	
4	8511	16731,000	0,000	0,000	L	Destilliertes Wasser (Aquadest)	0,000	

Staff Number 9999

Weigh comp.

Quantity adjustm.

Reject charge

Complete charge

04.07.14 10:39:51

A warning message the user may skip is output if the value of the actual deviation is greater than the one of the target deviation.

"Reject charge" function

Due to the weighing process, it might be the case that an entire batch cannot be used. The "reject charge" function can be used to identify the current charge/batch as "scrap". The material used for this charge is uploaded to ERP and the generated scrap quantity (identified as charge scrap by SYSTEM reason 910) is posted to the order. The batch is generated with the "locked" status and the "scrap" batch class (goods movement 531).

A confirmation prompt the user has to affirm comes up with this function.

Then the original target quantity for each charge is restored and the components are reset. Consequently, the weighing process can be restarted for a new batch.

"Adjust quantity" function

Once the first component has been weighed, the user can apply the "adjust quantity" function to automatically adjust the input quantities of the other components in proportion to the weighing result of the first component and its default quantity. This function can only be used once for each charge.

The function adjusts the target quantities of all components to the weighed quantity of one component. Quantities are adjusted by changing the primary target quantity for each charge of the operation.

However, quantities may only be adjusted if the component is weighed within its tolerances.

"Complete charge" function

Using this function, the charge/batch is completed (generated) and the dialog is closed. However, the charge/batch can only be completed, once all components have been weighed and are within their specified tolerances.

Completed successfully, the charge/batch is generated and transferred as goods movement 101 to ERP. Optionally, the charge/batch is assigned a minimum shelf-life (from the material type of the OP). In addition, the PPS batch from the order is determined and stored as PPS batch in the batch of the charge (optionally).

The quantity of the generated batch results from the input quantity recorded as consumption for each component.

The user may also directly block the batch/charge by selecting a status from the component requirements dialog. By default, the batch is always assigned the status "free".

34 Configuration of Weighing Components

Usage

You use the "pass batch attributes on" function if attributes of input batches are to be transferred to the generated output batch when changing output batches.

Dialog configuration

The input function is controlled by the dynamic dialogs A_VBRKOMP and KOMP_WIEG

Activation of the posting function at the terminal

Specific posting functions are enabled at AIP by an entry in the file **ctaipbut.ini**.

This is an example for the entry in ctwinbut.ini:

CTAIPBUT.INI
F1=A_VBRKOMP, weigh



The dynamic dialogs A_VBRKOMP and KOMP_WIEG must be available.

System configuration

Operation data

The following additional fields have be filled out for the operation using the PPS interface:

- No batch management requirement
- Target quantity per charge (calculated form primary target quantity / secondary quantity)(ab.soll_menge_ansatz)
- Number of charges (secondary quantity in pieces) - default = 1
- Batch

Data included in component list

These parameters have to be filled out for discrete material components using the PPS interface:

- Tolerance (in percent) → mlst_hy.mengen_tol
- Deviation (absolute value) → mlst_hy.mengen_abweichung

- Input quantity and unit of input quantity
- Component type must be D - discrete

35 Palletizing and Packing on the AIP

Log packing OP on

Use the input function (dialog A_AN_HU or A_P_AN_HU) to log on a packing OP requiring batch management to the terminal.

When logging the OP on, an operation is selected from the sequencing list and logged on to the machine. The logon is performed without input material for OPs requiring batch management and posted using the PDM command A_AN or A_P_AN.

If the machine is configured so that the person has to be logged on with the OP, the A_P_AN command is used and the dialog has to include an input field for the badge number.

Assign batches

The below input function (dialog CE_AN_HU) is used on the terminal to assign batches to the current TPU (running output batch) for a running OP requiring batch management:

Assign batches <CE_AN_HU>

Machine: MPDV-PAC Transp. unit: HUPBF9011D
Order: PLATE000320000850

Batch No.

Batches in transport unit

* Batch No.	Material	Weight	Einh	Designation
- PRPBF90116	PLATE_VIP07	1,00	KG	Plate finishing Article

No. of batches: 1 Total qty.: 1

Staff badge number:

Figure: "Assign batches" dialog - CE_AN_HU

Enter a batch number that is known in HYDRA and click the button *Assign batch* to logically assign the batch to the current TPU. Once assigned successfully, the batch is set to the "processed" status and displayed in the list, which includes all batches assigned to the TPU.

Note: You can assign only batches of the same material.

Use the button *Remove batch* to cancel the assignment of a selected batch (double-click on the relevant row). Then the batch changes back to the "free" status.

As additional status information on the current TPU, the dialog shows the number of assigned batches and their total quantity in kg.

You configure the contents of the "batches in transport unit" table in the section [C_PAL_ZUORD.LST] of the file ctaipplay.ini.

Complete TPU

You use this input function (dialog CA_WL_HU) to complete the currently running output batch (TPU) on the terminal.

Machine MPDV-PAC		Material PLATE_WIP07	
Order PLATE000320000850		New TPU HUPBF9P116	
BatchNo. 1			
Target buffer 0002			
Transp. unit SYSTEM			
Status ☒ Free ☐ Locked ☐ Waste	Tare weight 3 Net weight 15	Gross weight 18 KG Reason 0	
Staff badge number			

Buttons: Cancel Prod. TPU GEWICHT Generate TPU

Figure: "Complete TPU" dialog – CA_WL_HU

Using the function *Complete TPU*, you can complete a TPU and enter the weight and the status yield, locked, scrap. When you complete the TPU, the relevant quantity (net weight) is booked to the running output batch and completed and a goods receipt (movement type 101) is posted. At the same time, the next output batch is logged on to the OP.

You can collect the following data with this posting:

Target buffer

Material buffer where the TPU batch is posted. By default, this is the machine's output buffer.

Gross weight/tare weight/net weight

By default, the total (in kg) of the assigned single batches is entered as net weight. If entered manually, it is calculated as follows:

$$\text{Gross weight} = \text{Tare weight} + \text{Net weight}$$

The values "gross weight" and "tare weight" are additionally saved as batch attributes for the TPU batch:

Gross weight - batch attribute 301 (ATTR:301)

Tare weight - batch attribute 302 (ATTR:302)

Transp. unit (transport unit)

A transport unit existing in the system can be assigned to the TPU batch.

Note: the weight stored for the transport unit is currently not taken over as tare weight to the input dialog.

Status (free/locked/waste)

By using the status selection, you can assign a status to the generated TPU batch: status "free" (yield material), status "locked" (class: yield material) or status "waste" (class: scrap).

In case of scrap, the quantity is posted as scrap for the operation and a scrap reason is entered.

Number of batches

Shows the number of batches currently assigned to the TPU. The value is saved as batch attribute 201 (ATTR:201).

Once the input has been confirmed by clicking OK, you can optionally enter further batch attributes if the attributes have been defined for the material type of the operation.

Once the posting has been completed successfully, the TPU batch is generated as "merged batch" and the relevant quantity and the goods movement are posted for the operation.

Log OP off / Interrupt OP

The packing OP can be interrupted or logged off on the terminal using this input function (dialog A_UN_HU or A_AB_HU).

Once selected, the user can interrupt or finish the packing OP:

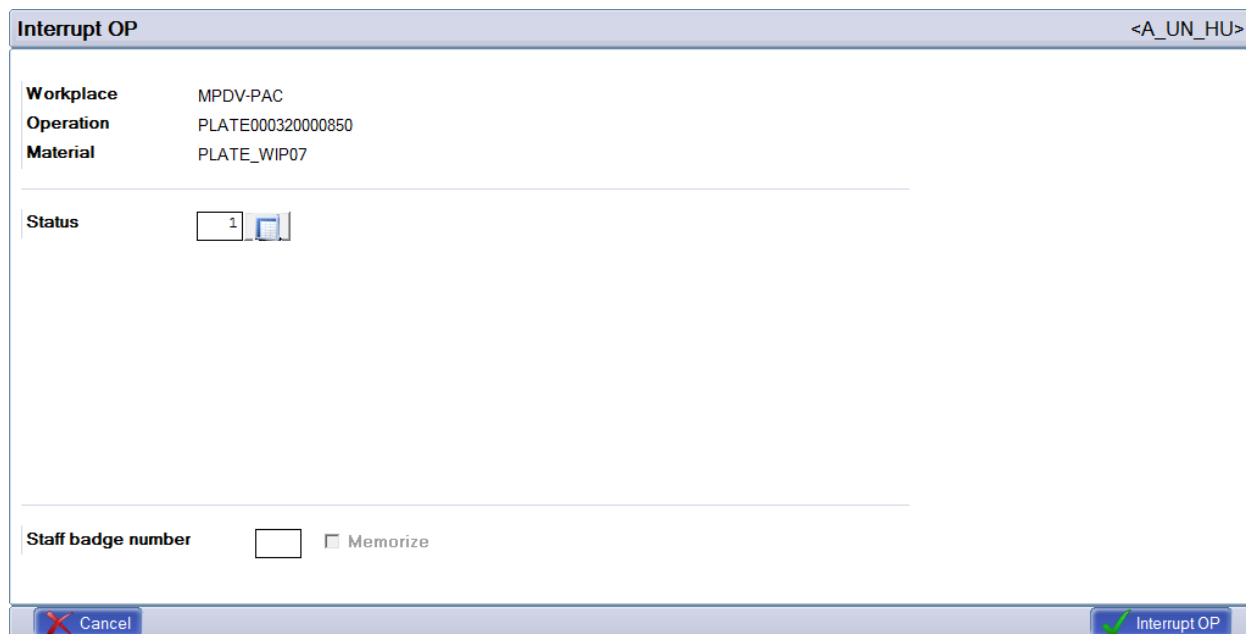


Figure: "Interrupt packing OP" dialog – A_UN_HU

The running packing OP is interrupted/finished after confirming by clicking OK.

Notes / restrictions:

- The last active output batch of the packing OP is completed without quantity and with status "deleted".
- If batches are still assigned to the TPU when you try to log the OP off, the logoff is rejected with error message.
- The ADE checks for overdelivery/underdelivery of the operation are not active when you use these posting functions.
- Quantities cannot be posted using this function.

Unpack / Repack

Requirements

Service pack 13 must be installed and activated. The document [Activating MPL TRT Dialogs \(SP13\)](#) describes how to use the dialog on the shop floor client.



For new customers, the new/updated MPL or TRT dialogs are directly available once the service pack 13 has been released.

General

Using this dialog, the operator can enter the packed handling unit and unpack selected batches. The operator can also unpack all batches and therefore cancel the handling unit.

Terminal procedure - Unpack individual batches

You start the function "Unpack TPU" (Dialog: CE_DEL_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit. You can double-click specific batches to select the batches that you want to unpack. The selected batches are labeled in the list with an "x". Double-click a selected batch to deselect the batch. The display returns to normal ("-").

Use the button *Unpack selected ones* to unpack selected batches.

A confirmation prompt opens asking whether you want to execute the dialog:

- Click *No* to cancel execution. The user returns to the dialog. The selected batches remain.
- Click *OK* to execute the dialog.

Then a message confirms that the unpacking process has been completed successfully. If you click *OK*, you get back to the dialog for further processing.

If you click *Exit*, the dialog closes. Selected unpacked batches remain unchanged in the handling unit.

Terminal procedure - Unpack all batches

You start the function "Unpack TPU" (Dialog: CE_DEL_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit. You can double-click specific batches to select the batches that you want to unpack. The selected batches are labeled in the list with an "x". Double-click a selected batch to deselect the batch. The display returns to normal ("-").

The function key *Unpack all* unpacks all batches contained in the handling unit.

Then a message confirms that the unpacking process has been completed successfully. If you click *OK*, the dialog is closed because there are no more batches.

Terminal procedure - Add new batch

You start the function "Unpack TPU" (Dialog: CE_DEL_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit.

If you click the button *Add*, you can add batches to the handling unit. A dialog opens where you can enter the batch number that is added, either manually or by scanning. After confirmation, the batch is added to the handling unit.

Posting procedure - Unpack individual batches

If you want to remove child batches from a handling unit, the system checks the following:

- The child batches must exist, but can be archived.
- The handling unit must be available but not running or processed.

The following actions are performed for each unpacked child batch:

- The batch status of the child batch is set to the status of the handling unit.
- The remaining quantity of the child batch is calculated from the total quantity of the goods movements when the child batch is packed.
- An entry in the [Batch history](#) is generated.
- The connection between handling unit and child batch is removed
- The batch assignment between child batch and handling unit is removed.

Actions performed for the handling unit:

- The remaining quantity of the handling unit is reduced by the sum of all remaining quantities of all unpacked batches.

- If the remaining quantity of the handling unit is zero, the batch is set to status "A" (processed)
- A goods movement is generated specifying the reduced quantity.

Posting procedure - Unpack all batches

See above "Unpack individual batches". The same processing is applied. All batches are unpacked and the handling unit with the remaining quantity = 0 is set to status "A" (processed).

Posting procedure - Add new batch

If you want to add child batches to an existing handling unit, the system checks the following:

- The handling unit must be available but not running or processed.
- The batch to be added must have the status "F" (free) and the same article number as the handling unit batch.

Note for the added child batch:

- All batch data of the handling unit is transferred to the child batch.
- The batch class of the child batch is identical to the batch class of the handling unit.
- The batch status is "A" (processed).
- The child batch is linked to the handling unit.
- An event „CE_AN_PA“ is added for the child batch in the batch history.
- The batch assignment between child batch and handling unit is generated.
- A goods movement (goods issue) is generated for the child batch.

Actions performed for the handling unit:

- The remaining quantity and the quantity of the handling unit increase by the remaining quantity of the added child batch.
- The system generates a goods movement (goods receipt) with the new quantity of the handling unit.

36 Activating Palletizing/ Packaging/ Assembling

Requirements

Ensure that:

- The flag "Batch management requirement" is set at the operation
- The operation material type does not have the flag "Consumption balance"
- The primary quantity unit for the operation is "KG"

Ensure that at the [workplace](#):

- The flag "Batch administration" is set
- The machine type "Packing location" is selected
- Machine monitoring is not activated

Procedure

Assign the [workplace to a terminal](#). Reboot the terminal.

Results

The following specific dialogs are available at the terminal for the configured packing location:

Dialog	Description
A_AN_HU	Log packing OP on (no input batches)
A_P_AN_HU	Log packing OP on (no input batches, with person)
CE_AN_HU	Assign batch to running OP/ TPU
CA_WL_HU	Complete TPU (output batch change)
A_UN_HU	Interrupt packing OP
A_AB_HU	Finish packing OP
A_AUT_HU	Log off/ interrupt OP selection
C_VLOS_HU	Display preceding TPU



The TPU batch numbers created are permanently generated at the terminal with the prefix HU.
The prefix is not currently configurable.



There is no off-line entry capability

37 Consumption Balance

Summary

When logging an OP off, the “consumption balance” (V_BLZ dialog) can be displayed, which can be configured via the machine and the material type of the operation.

The consumption balance is shown if this option is active at the machine and material type.

The consumption balance shows the material consumption based on batches and the user is able to log still running batches off.

Dialog

Verbrauchsbilanz anzeigen <V_BLZ>

Arbeitsplatz: MPL301 Artikel: 10-20
Arbeitsgang: T00000010020

Gesamtmenge: 272 ST
Menge: 25 ST

POS	Material	Soll-Bedarf	Th.Verbrauch	Ist-Verbrauch	Toleranz	Mat.Typ	Los
1	10-10	1000,000	297,000	0,000 ST	0,000	SYSTEM	WE9LD7A142
2	99-99	2000,000	594,000	0,000 ST	0,000	SYSTEM	WE9LD7A1A2

Personalkartenummer: 0001

Abbrechen Details anzeigen Los abmelden OK

Figure: Consumption balance – V_BLZ

Display of consumption postings

The “show details” function allows for consumption quantities, which have been collected so far, to be displayed.

Detail Verbrauchsbilanz anzeigen
<V_BLZ_DTL>

Arbeitsgang T00000010020

Artikel 10-10

Artikelbezeichnung

Personalkartennummer 0001

Errechnete Menge 297 ST

Gemeldete Menge 90 ST

POS	Losnummer	Verbrauch	Anmeldedat	Anmeldezeit	Abmeldedat	Abmeldezeit	STA
1	WE9LD7A141	90,000 ST	09.08.2010	16:53:58	10.08.2010	09:22:58	F

Figure: Display of consumption postings – V_BLZ_DTL

Logging input batches off

By way of the “log batch off” function, the user can choose a currently running batch from the list and log it off by entering consumption.

Eingangslos abmelden		<V_BLZ_CEAB>	
Los	WE9LD7A142	Material	10-10
Stücklistenposition	1	Materialkurztext	
		Materialtyp	SYSTEM
Status ☐ ABGEARBEITET ☐ GESPERRT ☒ mit Restmenge		Verbrauch 297 ST	Loshinweis mg 100000
Personalkartennummer		0001	
Abbrechen		Eingangslos abmelden	

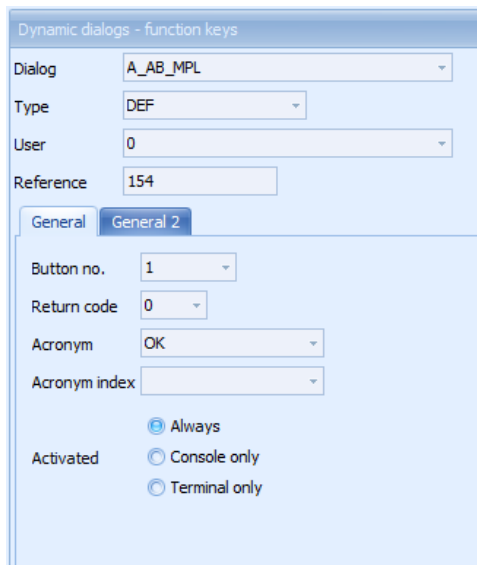
Figure: Log batch off – V_BLZ_CEAB

38 Configuration of Consumption Balance

Dialog configuration:

A special configuration (function = "DLG=V_BLZ;BREAK-ON-CANCEL") has to be defined for the OK button of the logoff dialog (e.g. A_AB_MPL) to make sure that the consumption balance is started when the OP is logged off.

Sample configuration of the OK button for starting the consumption balance:



Dynamic dialogs - function keys

Dialog: A_AB_MPL

Type: DEF

User: 0

Reference: 154

General

General 2

Button no.: 1

Return code: 0

Acronym: OK

Acronym index:

Activated: ☒ Always ☐ Console only ☐ Terminal only

Dynamic dialogs - function keys

Dialog: A_AB_MPL

Type: DEF

User: 0

Reference: 154

General General 2

Key:

Text: OK

X pos.: 505

Y pos.: 12

Width: 105

Height: 30

Information: Uebernehmen

Symbol: UPD.BMP

Function: DLG=V_BLZ;BREAK-ON-CANCEL

License:

User-def. 1: AUTO

☐ Blocked

User-def. 2: FORM-VALIDATION

Figure: Example for the OK button configuration including consumption balance